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CND Lab Report (Volume 3 of 3)

EC-Council CND (Certified Network Defender)

Data Security – File/Partition Recovery & Disk Encryption (BitLocker)

EXECUTIVE SUMMARY

Labs 8 through 11 walks through the full **data-resilience lifecycle** on isolated virtual test environments, deliberately never touching production/system drives:

- **Lab 8 (Kernel for Windows Data Recovery)** file-level detection on the C: drive; recovery blocked by a trial license, but detection accuracy fully verified.
- **Lab 9 (MiniTool Power Data Recovery)** partition-level recovery: an entire deleted partition was located and a file inside it recovered.
- **Lab 10 (BitLocker)** the "prevention" counterpart: full-disk encryption (XTS-AES 128) with password + recovery-key protection on a dedicated non-system drive.
- **Lab 11 (Disk Drill)** file-level recovery after Recycle Bin emptying, using deep scanning to restore a document and an archive with verified content.

Together, three tools demonstrate that **deletion is reversible** at both the file and partition level until the underlying disk space is overwritten, while the fourth (BitLocker) demonstrates the **preventive** control that makes data unreadable even if physically accessed.

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File Recovery Using Kernel for Windows Data Recovery

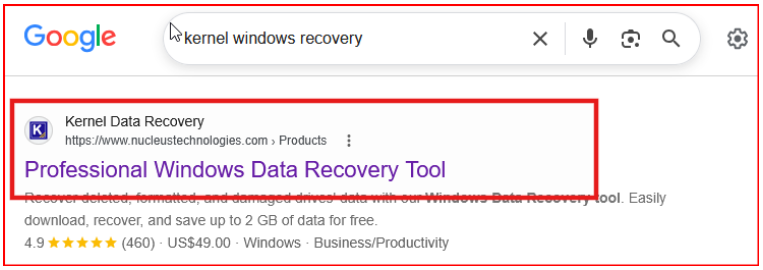
Executive Summary

In this lab you create sample documents, permanently delete them, scan the C: drive, attempt to recover the deleted folders/documents, and verify that the recovery tool correctly located the deleted data. Unlike Labs 9–11, this lab does not end in a full recovery — the trial version blocks the final save step — but it does successfully validate detection.

Lab Requirement	Details
Virtual machine	AdminMachine-1
Tool	Kernel for Windows Data Recovery
Objective	Recover permanently deleted folders and documents
Scan location	Local Disk (C:)
Recovery destination	Desktop → Recovered Data
Expected result	Deleted folders and documents restored successfully

Install the Recovery Tool

Kernel for Windows Data Recovery official download

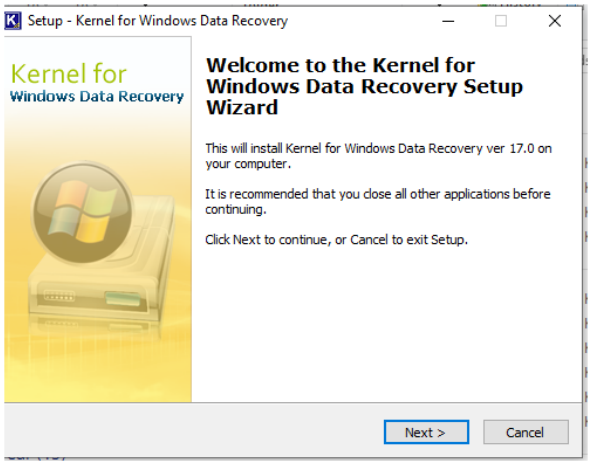


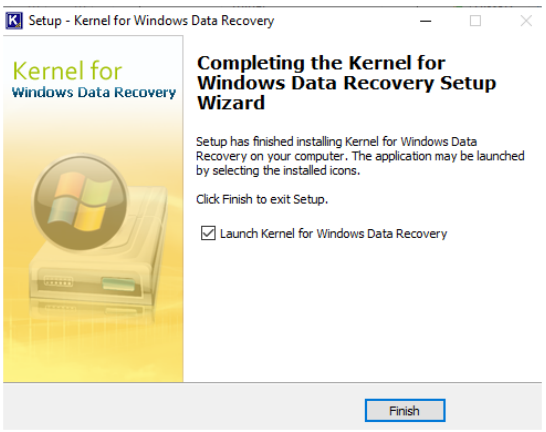
Take free trial



Launch Kernel for windows data recovery

Click next



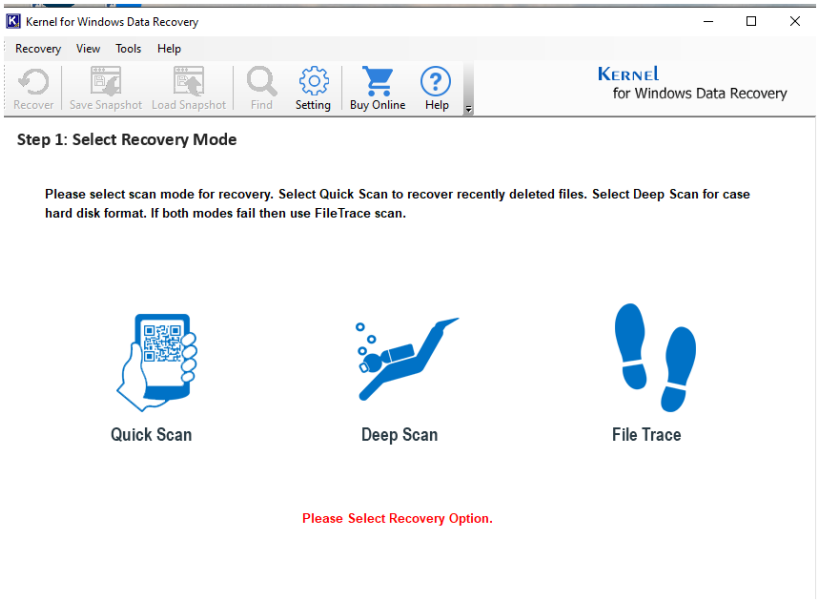


Click Try now



Create the Test Data

- On the AdminMachine-1 desktop, create two folders: Test Data and Recovered
- Inside Test Data, create two text files: Employee Records.txt and Security Notes.txt
- Enter content into each:
    - Employee Records.txt → "Employee records created for the Windows data recovery lab."
    - Security Notes.txt → "Security notes created for testing deleted-file recovery."
- Save and close both files



Create the Test Data

1. On the AdminMachine-1 desktop, create two folders:

Test Data

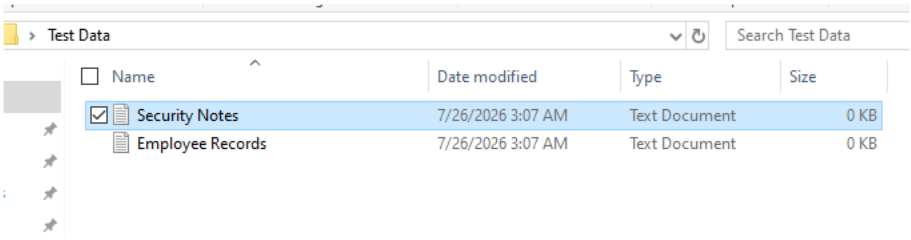
Recovered



3. Open **Test Data**.
4. Create two text files:

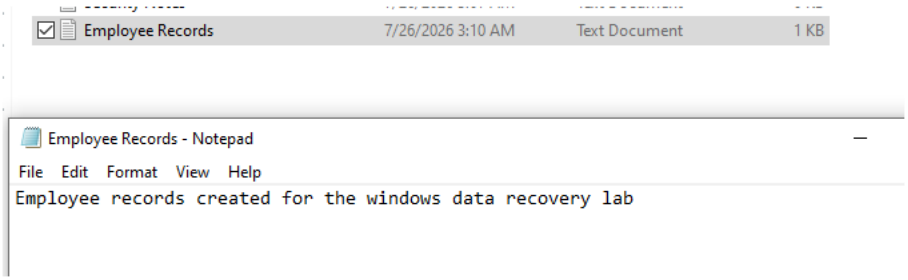
Employee Records.txt

Security Notes.txt



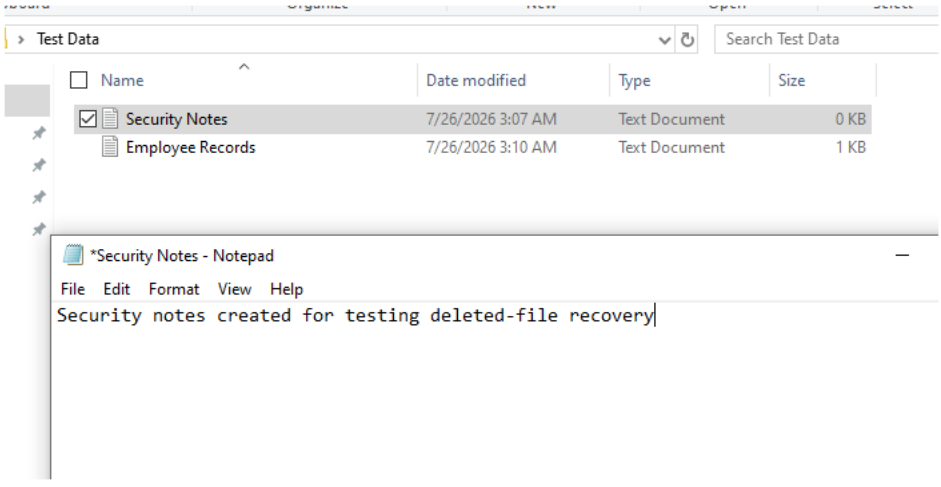
5. Open **Employee Records.txt** and enter:

Employee records created for the Windows data recovery lab.



6. Open **Security Notes.txt** and enter:

Security notes created for testing deleted-file recovery.



7. Save and close both files.

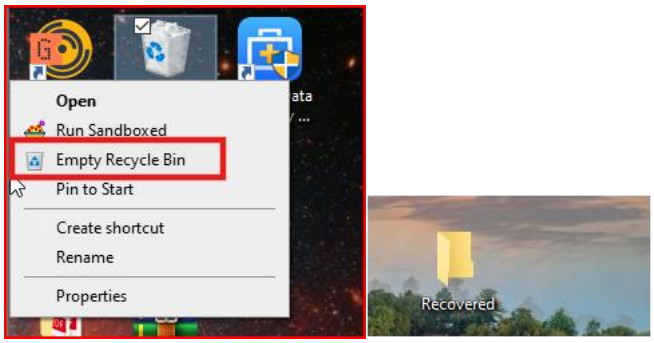
Permanently Delete the Test Data

- Delete the Test Data folder (leaving Recovered in place)
- Empty the Recycle Bin to force permanent deletion

Before

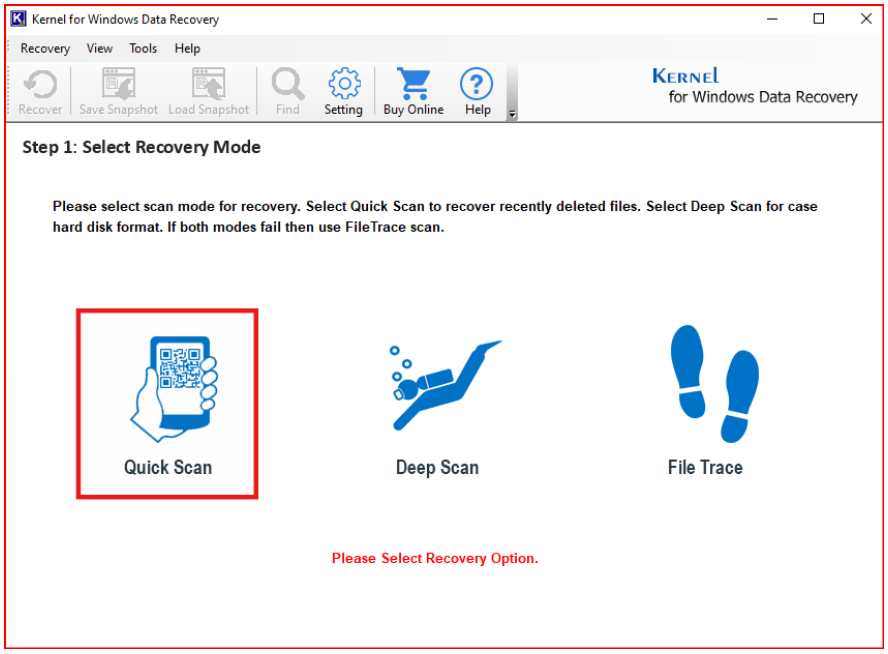


After



Scan the C: Drive

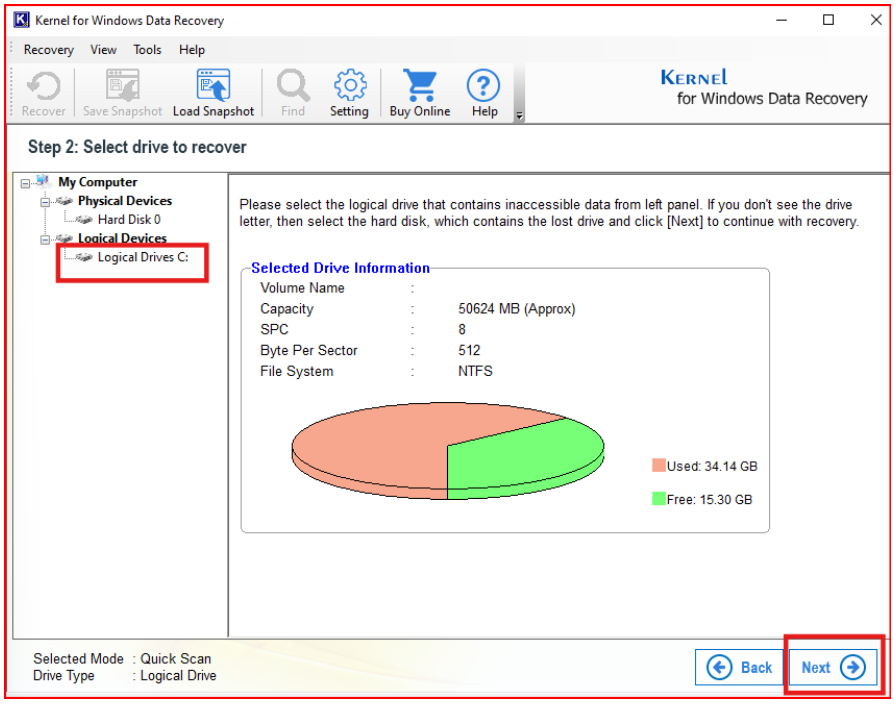
- Return to Kernel for Windows Data Recovery, click Try Now if the registration window reopens
- Select Quick Scan as the recovery mode
- Select Local Disk (C:) as the drive to recover
- Click Next / Start Scan and wait for the scan to complete

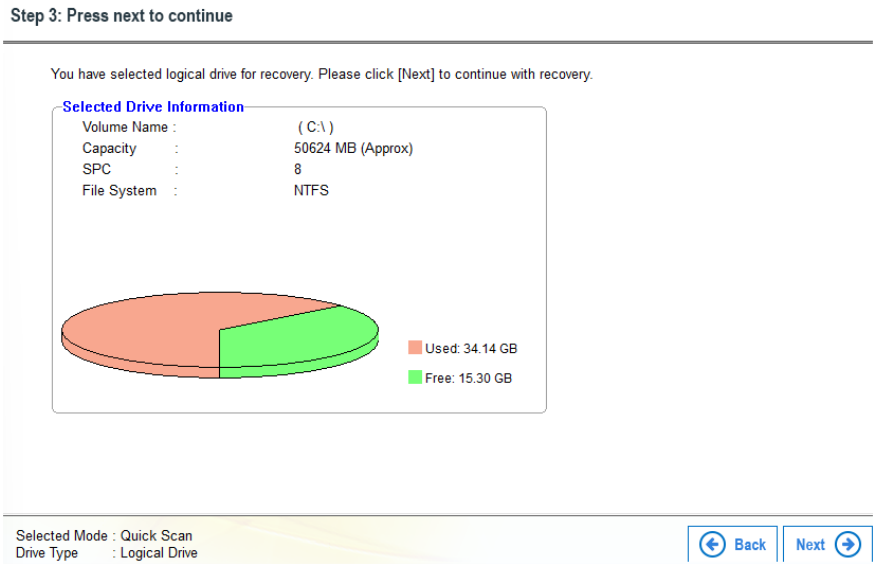


1. Select:

Local Disk (C:)

Click **Next** or **Start Scan**.

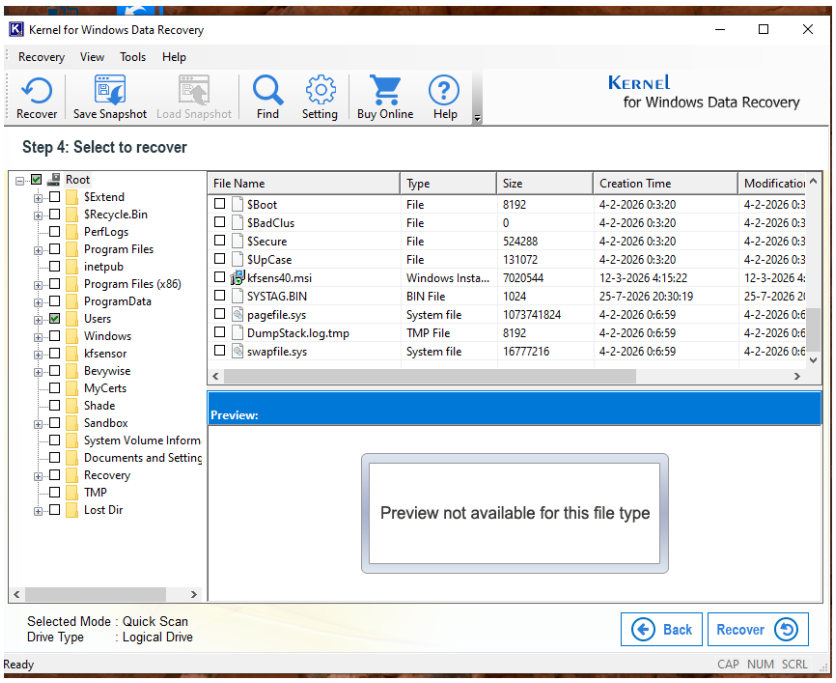




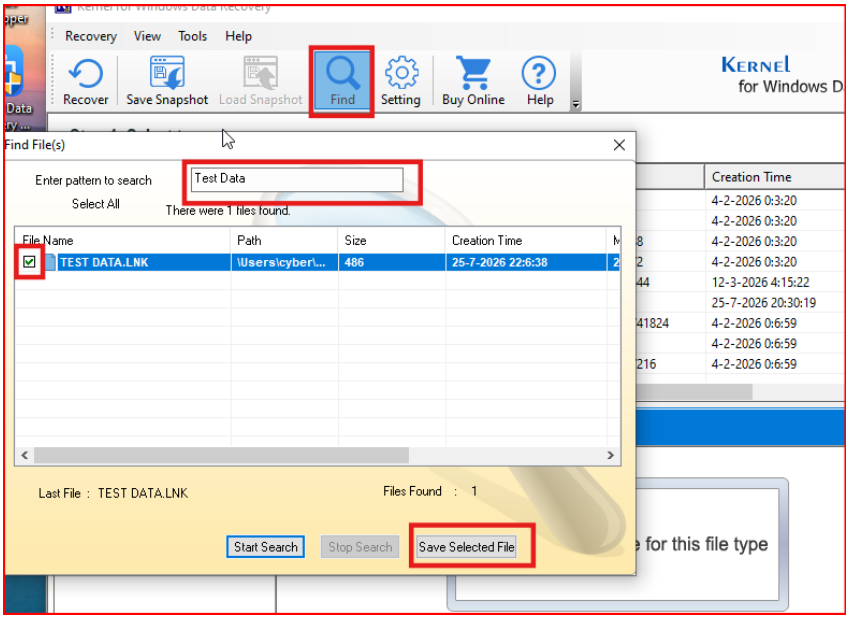
Locate the Deleted Files

- In the scan results tree, use the Find tool and search the pattern "Test Data"
- The search returns TEST DATA.LNK (486 bytes)
- Select the file and attempt Save Selected File

In the scan results:



Find Test Data -> the file we deleted before

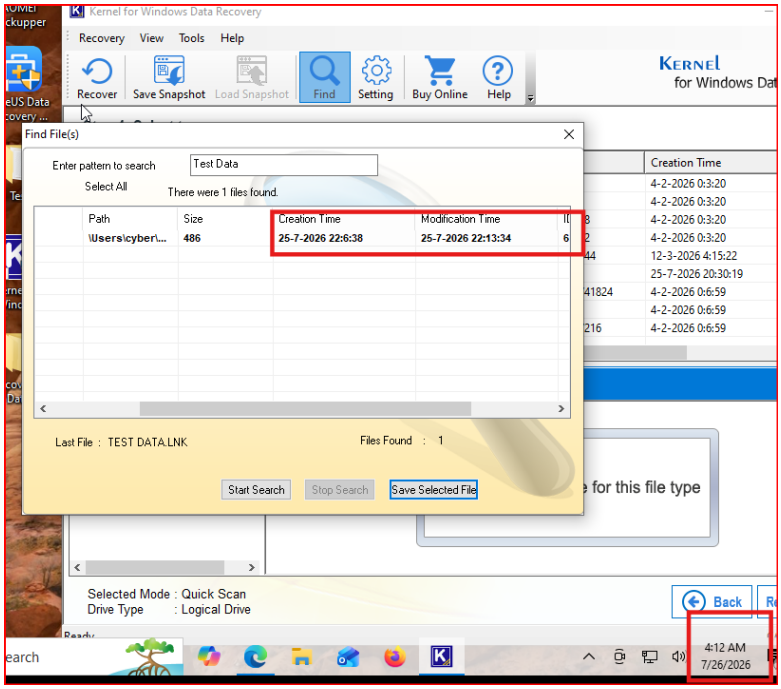


When we click the save selected file it ask for premium access



Verification & Licensing Block

- Saving triggers a registration prompt requiring Buy Now / Activate Now (premium license)
- Cross-check the file's **Creation Time** and **Modification Time** against the actual deletion date/time (25-7-2026, matching system clock 7/26/2026)—timestamps match, confirming the correct deleted item was found
- Full file save is blocked because the trial edition requires a paid license



Conclusion

Lab 8 confirms that Kernel for Windows Data Recovery can detect permanently deleted files and folders on the C: drive with accurate metadata (filename, creation date, modification time), validating that deletion does not immediately erase data. However, the trial version could not complete the final recovery (save) step — that capability is gated behind a premium license.

Final Lab Result: Kernel for Windows Data Recovery successfully detected and displayed the deleted test files. File details matched the original filenames, deletion date, and system time. Saving the recovered files was unavailable because the trial version required a premium license.

Partition Recovery Using MiniTool Power Data Recovery

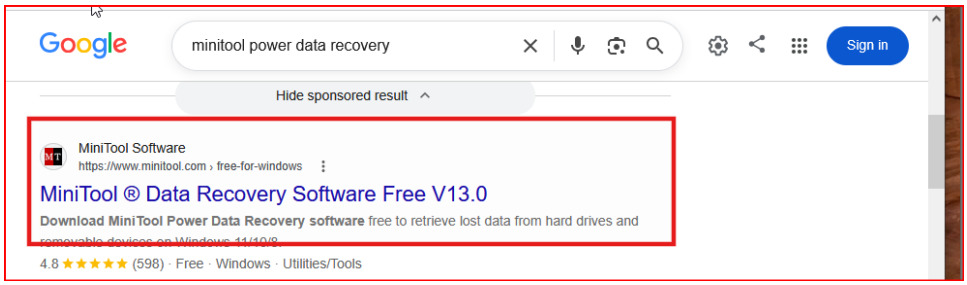
Executive Summary

In this lab you use MiniTool Power Data Recovery to build a small virtual test partition, delete it entirely, scan for the lost partition, and recover the test file from inside it — validating partition-level recovery rather than single-file recovery.

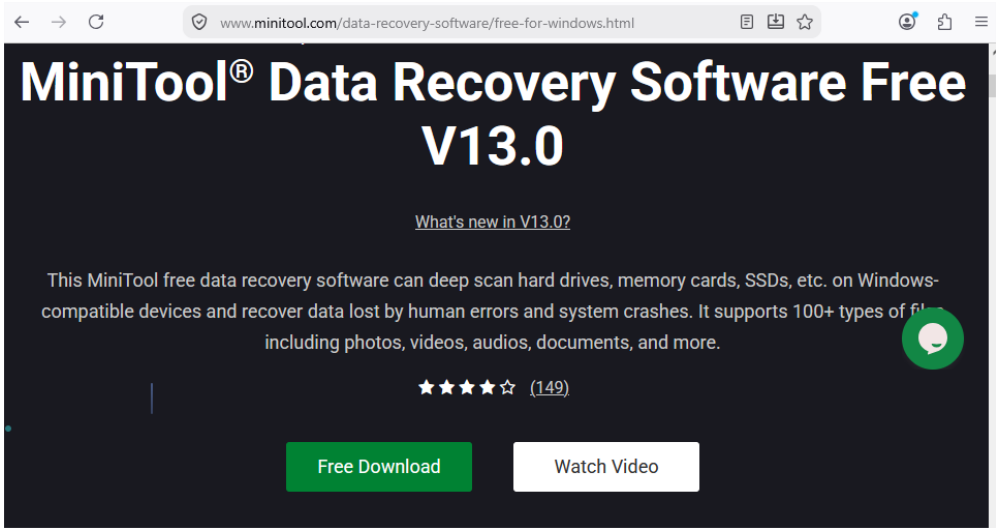
Item	Details
Virtual machine	AdminMachine-1
Objective	Recover deleted data from a partition
Required tool	MiniTool Power Data Recovery Free
Expected result	Deleted files are detected and recovered

Install MiniTool Power Data Recovery

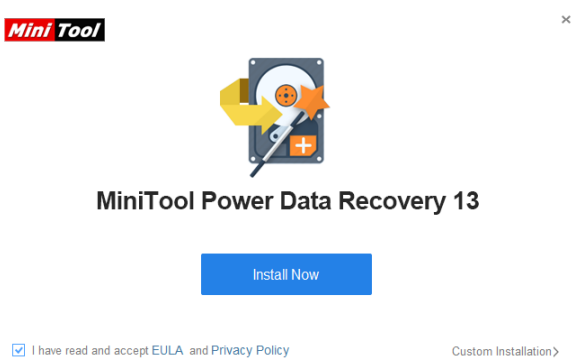
- Search "minitool power data recovery," open the official site, click Free Download → Install Now

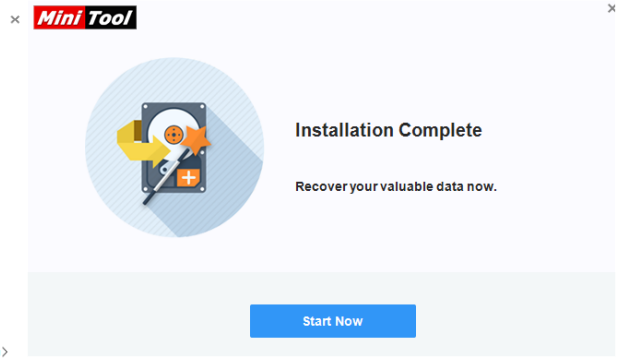


Click to download it for free

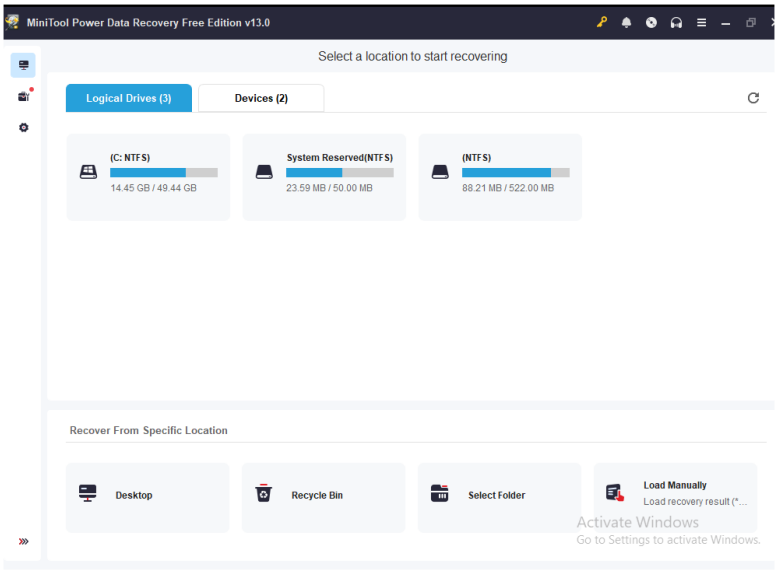


Click install now





Launch MiniTool Power Data Recovery

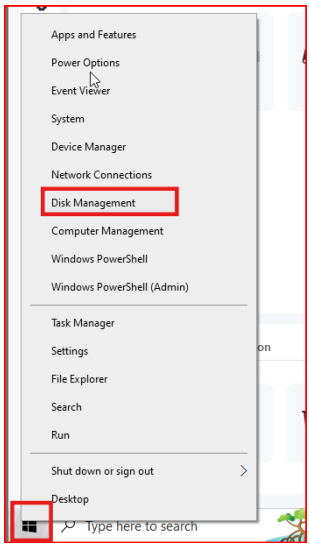


Creating a Safe Virtual Test Disk

- **Open Disk Management → Action → Create VHD**
- **Location: C:\Lab9-TestDisk.vhdx, Size: 512 MB, Format: VHDX, Type: Dynamically expanding**
- **New disk appears as Unknown / Not Initialized / 512 MB → Initialize Disk → MBR**

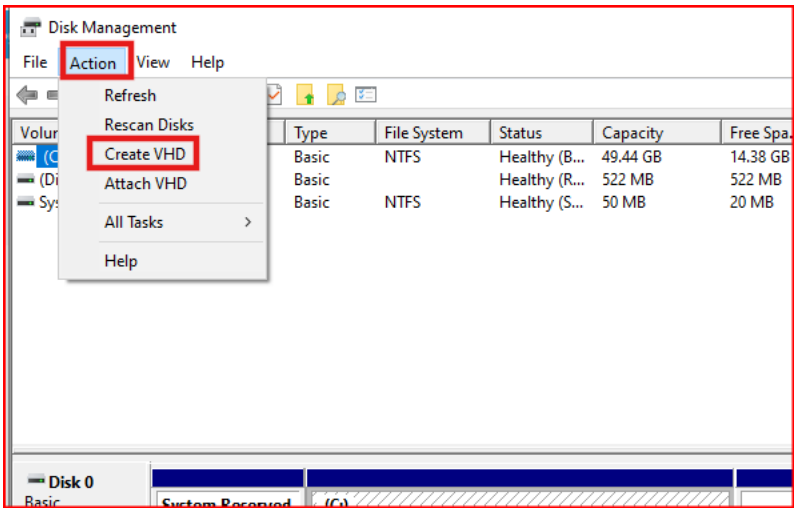
Right-click the Windows **Start** button and select:

Disk Management



3. In Disk Management, click:

Action → Create VHD



4. Configure the virtual hard disk:

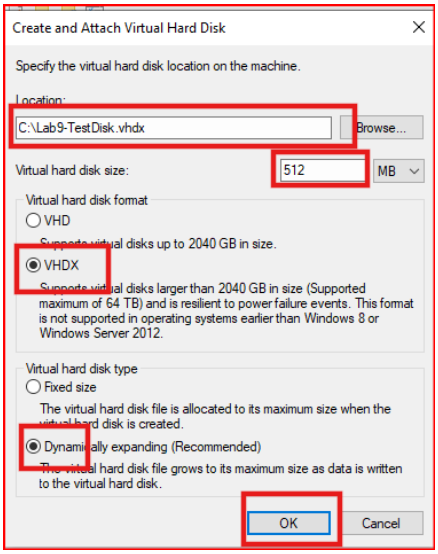
Location: C:\Lab9-TestDisk.vhdx

Virtual hard disk size: 512 MB

Format: VHDX

Type: Dynamically expanding

5. Click **OK**.

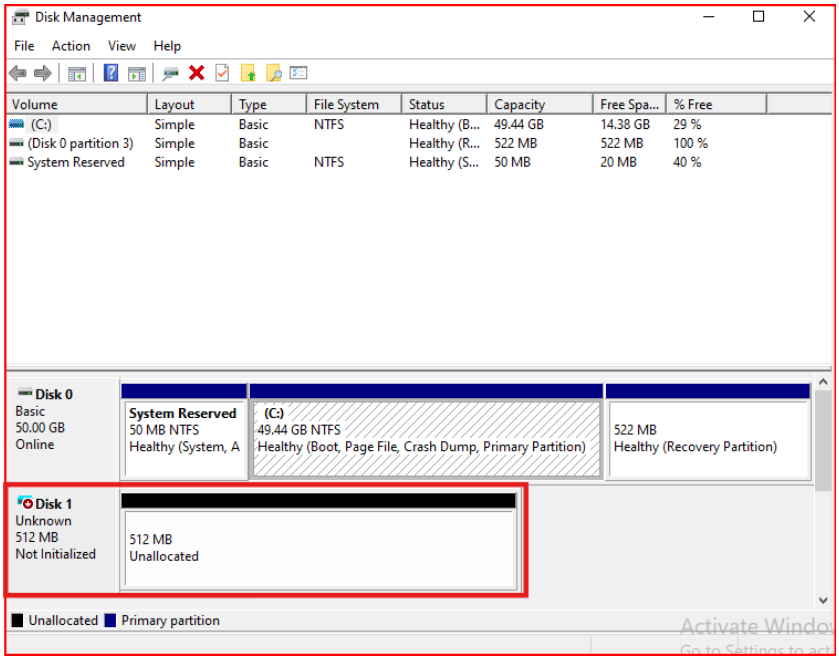


5. A new disk will appear at the bottom as:

Unknown

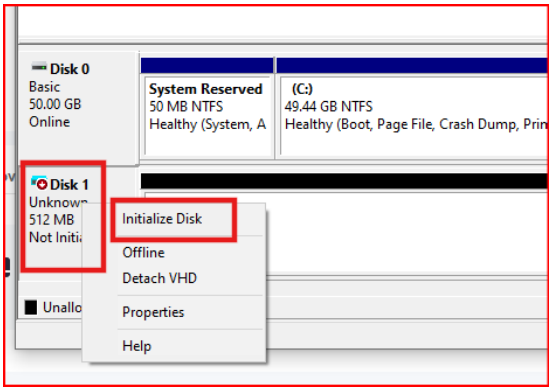
Not Initialized

512 MB



7. Right-click the left section containing **Disk 1** or the newly created disk number.
8. Select:

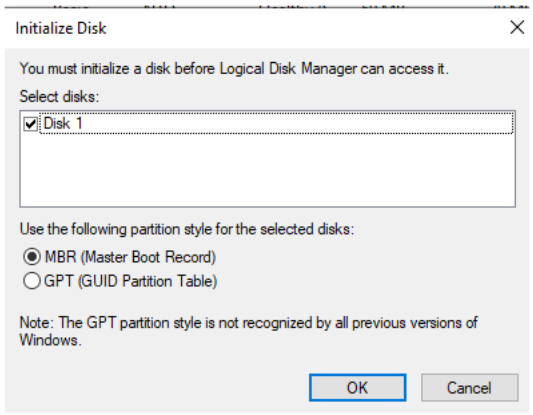
Initialize Disk



9. Choose:

MBR

10. Click **OK**.



Create the Test Partition

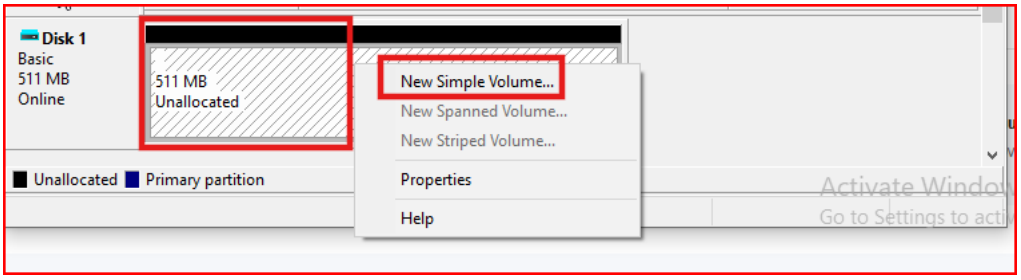
Right-click the 512 MB Unallocated space → New Simple Volume

Assign drive letter (e.g., A:), File system: NTFS, Volume label: `Lab9 Test`, Quick Format checked

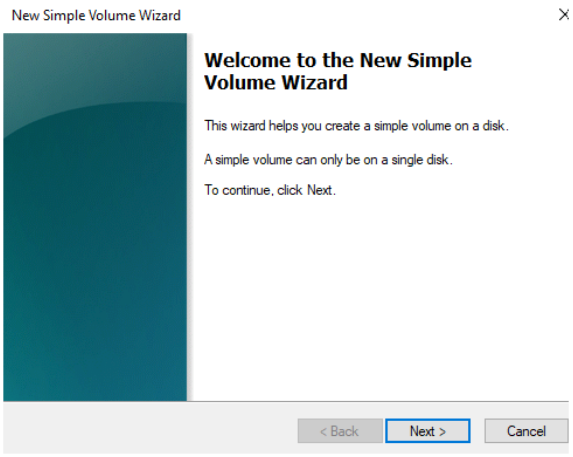
Confirm File Explorer shows **Lab9 Test (A:)**

Right-click the **512 MB Unallocated** area. -> Select:

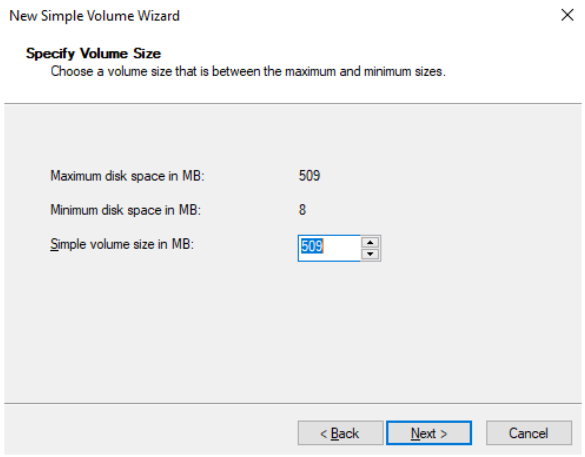
New Simple Volume



Click next

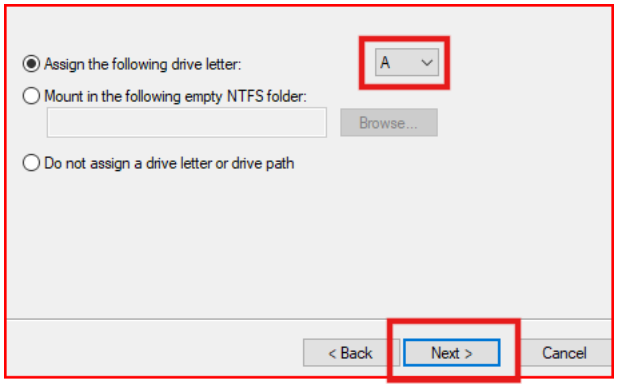


Click Next (default)



3. Assign an available drive letter, such as:

A:



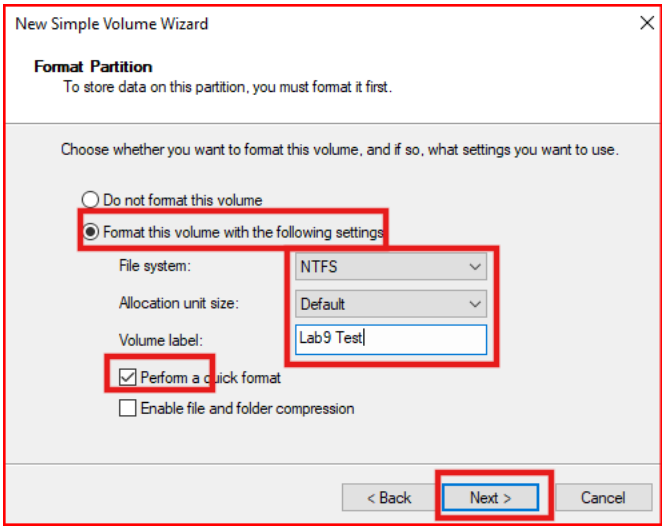
5. Select:

File system: NTFS

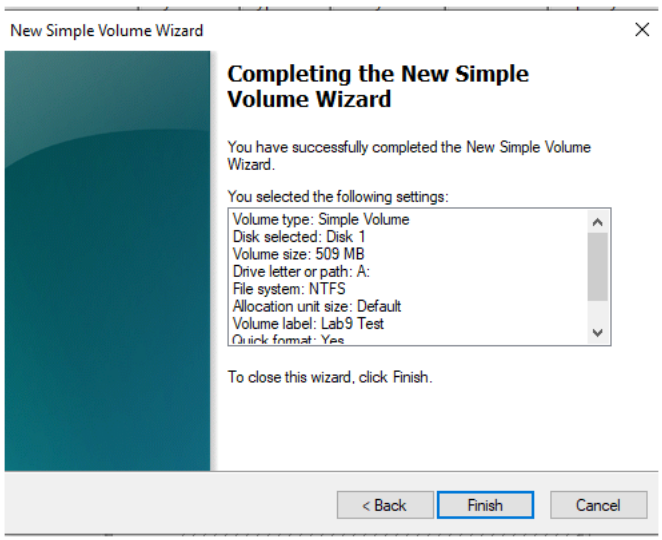
Allocation unit size: Default

Volume label: Lab9 Test

Perform a quick format: Checked



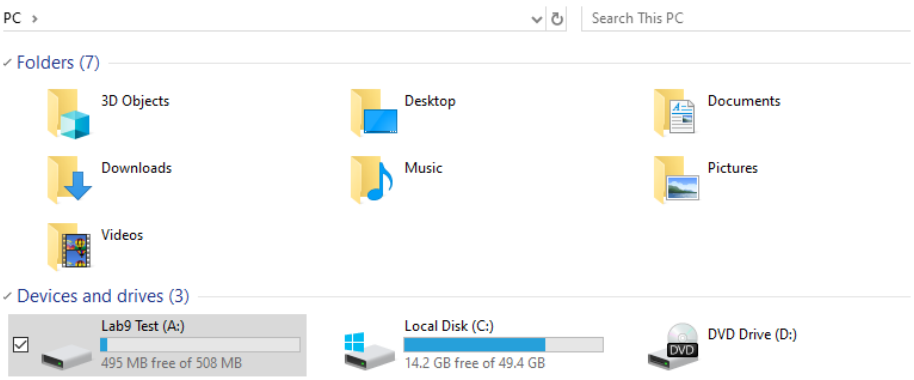
6. Click **Next → Finish**.



Expected Result

File Explorer should now show:

Lab9 Test (A:)



Add Test Data

Create `PR Test.txt` on the new partition, content: "Test data added"

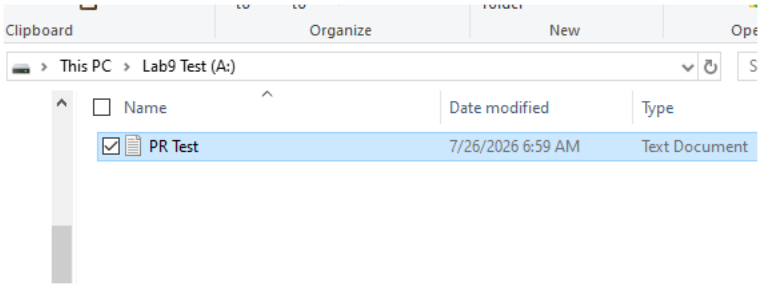
Create the recovery destination folder on the desktop: `Lab9 Recovered`

Open:

This PC → Lab9 Test (D:)

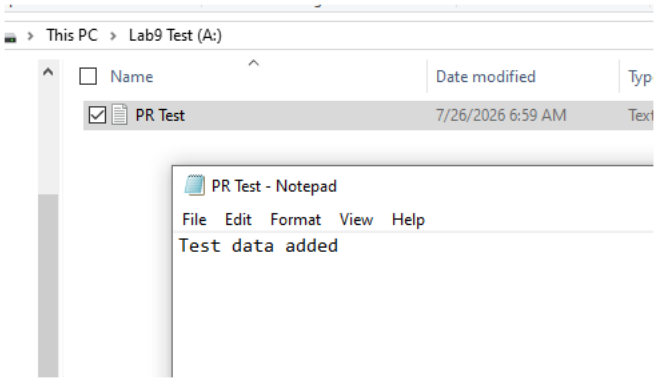
2. Create a new text document named:

PR Test.txt



3. Open the document and enter:

Test data added



4. Save and close the document.
5. Create this recovery destination on the desktop:

Lab9 Recovered



Delete the Test Partition

In Disk Management, right-click the Lab9 Test (511 MB) partition → Delete Volume

Confirm the space now shows as **Unallocated** (do not detach the virtual disk)

Return to **Disk Management**.

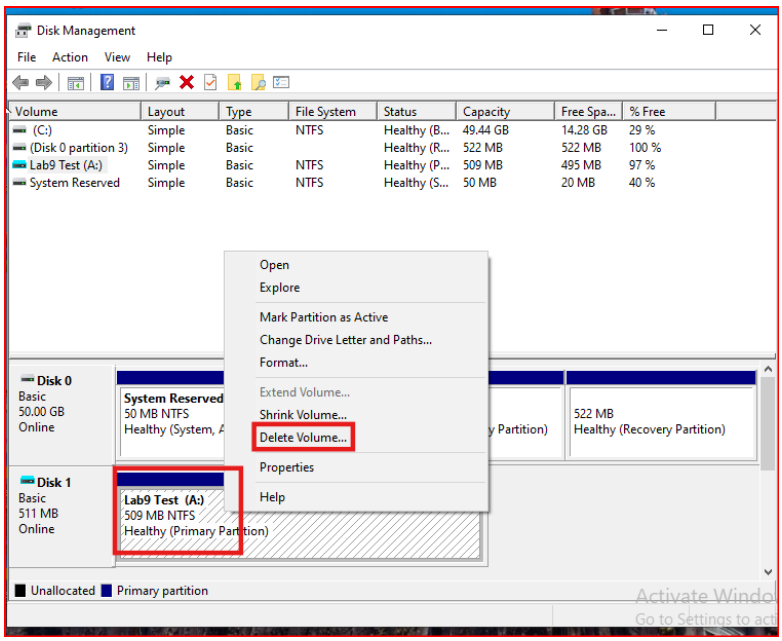
Locate the partition labeled:

Lab9 Test

512 MB

3. Carefully right-click the blue partition area for **Lab9 Test**.
4. Select:

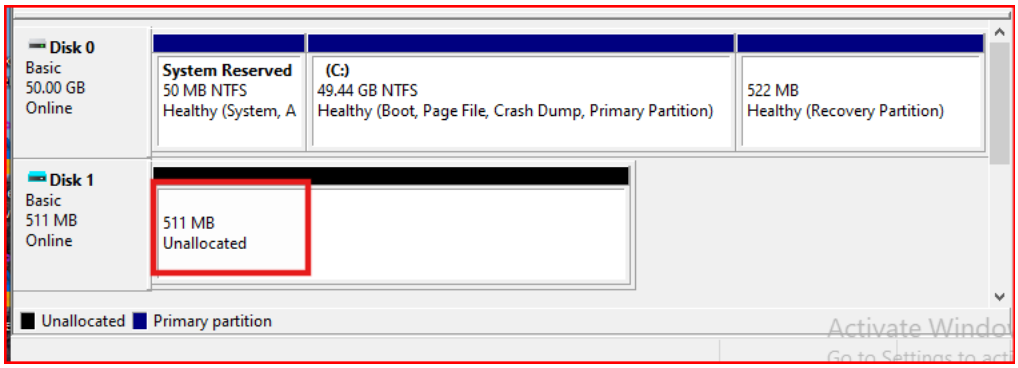
Delete Volume



Expected Result

The 512 MB virtual disk should now display:

Unallocated



Do not detach or delete the virtual disk.

Your **511 MB test partition is correctly deleted and shown as Unallocated**. ✅

Scan the Lost Partition

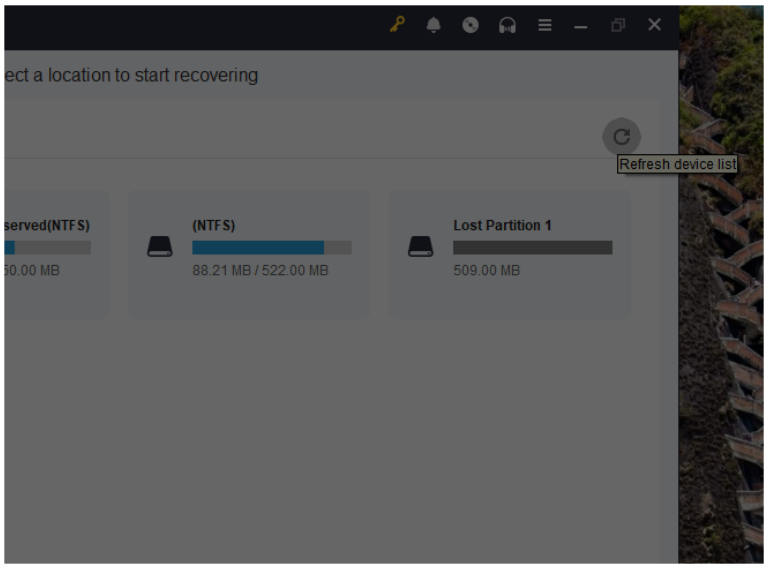
- Return to MiniTool Power Data Recovery, click Refresh
- Under Logical Drives, locate Lost Partition 1 (511 MB), hover and click Scan

- Wait for the scan to reach 100% (do not scan C:, System Reserved, or the recovery partition)

Close **Disk Management**.

Return to **MiniTool Power Data Recovery**.

Click the **Refresh** icon in the upper-right corner.



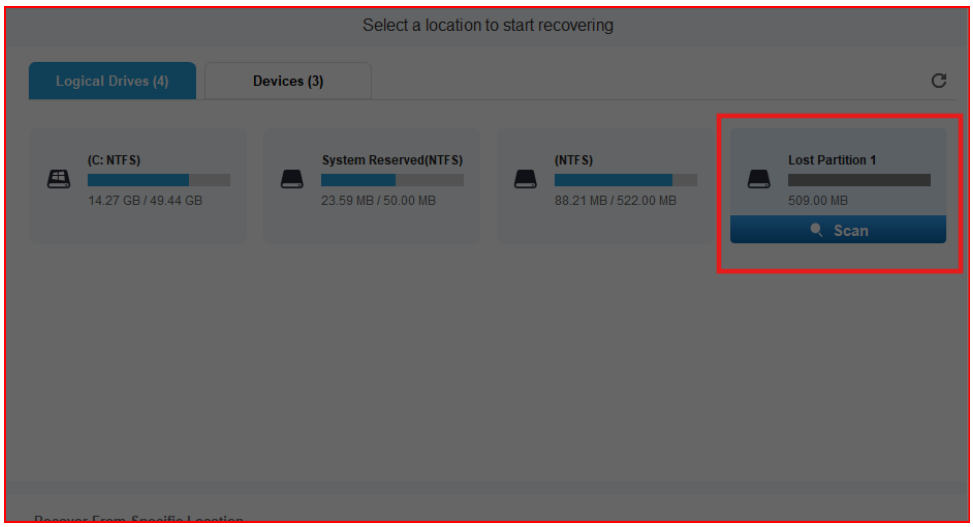
1. Under **Logical Drives**, locate an entry similar to:

Lost Partition 1

511 MB

5. Move the mouse over **Lost Partition 1**.
6. Click:

Scan



Note: Do not scan the C: System Reserved, or 522 MB recovery partition.

Wait until the scan reaches 100%.

Locate and Recover the Test File

Open the Path tab, expand the detected partition, locate `PR Test.txt`

Select its checkbox; Preview is unavailable in the free trial without downloading a previewer, so save instead

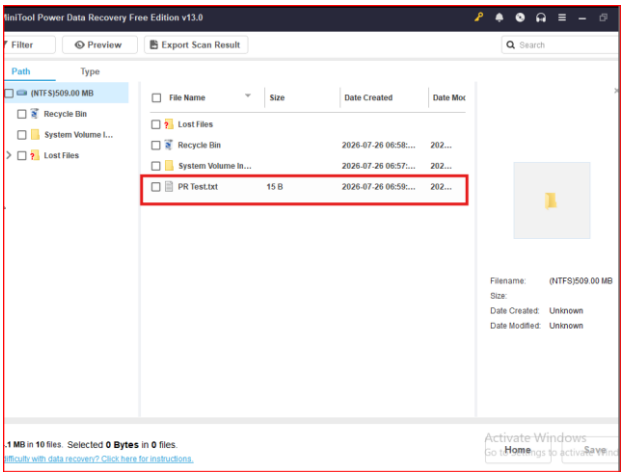
Choose destination: Desktop → `Lab9 Recovered`, click OK

Open the **Path** tab on the left.

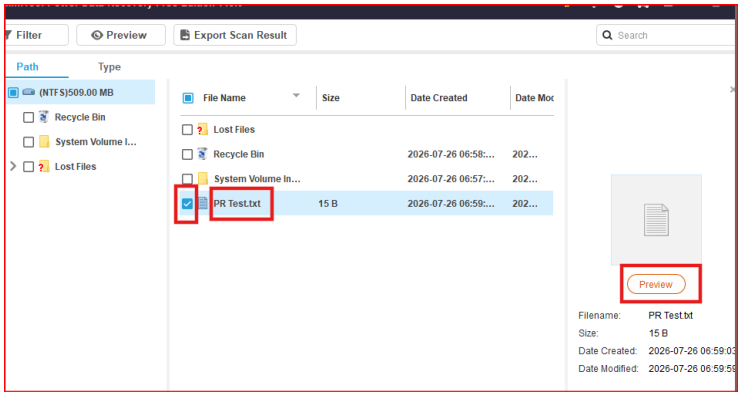
Expand the detected partition or folders.

Look for:

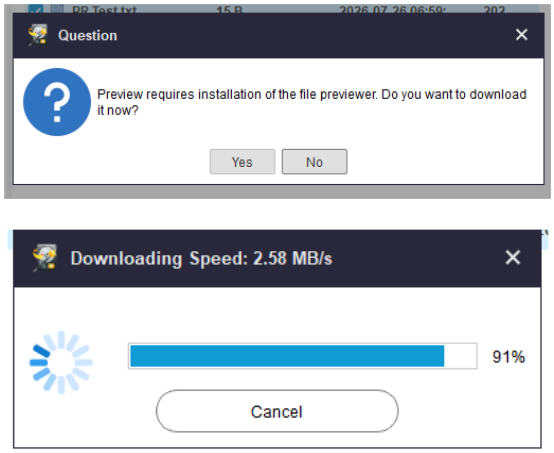
PR Test.txt



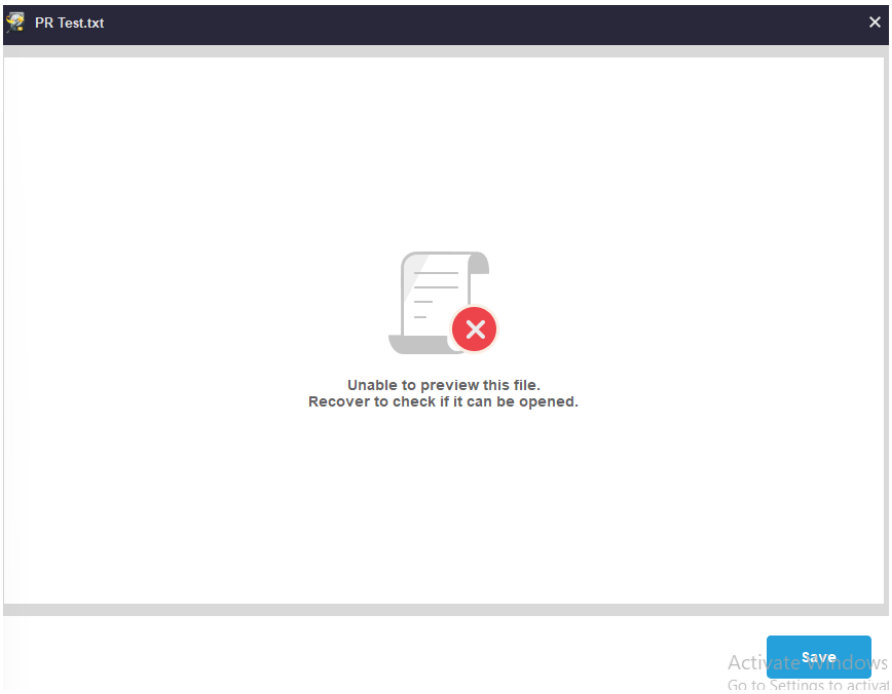
4. select the checkbox beside the file.
5. Click **Preview** if available.
6. Confirm that the preview contains:



Select Yes



Unable to preview

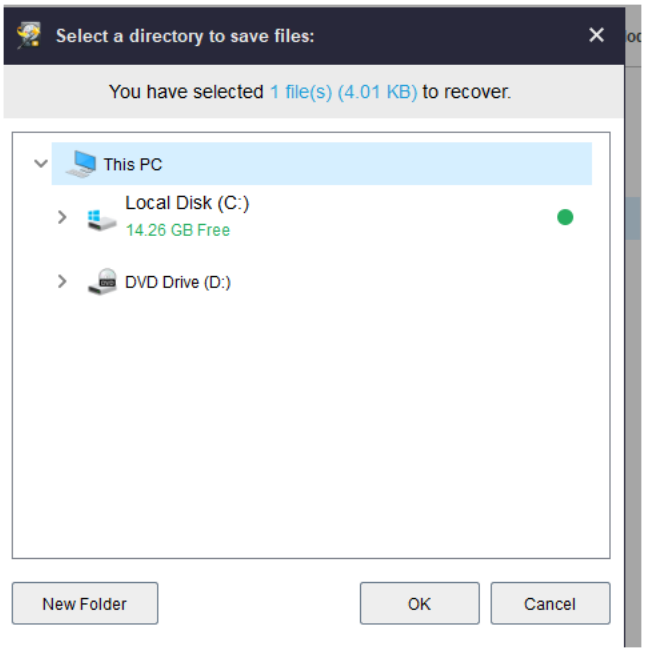


Verify Recovery

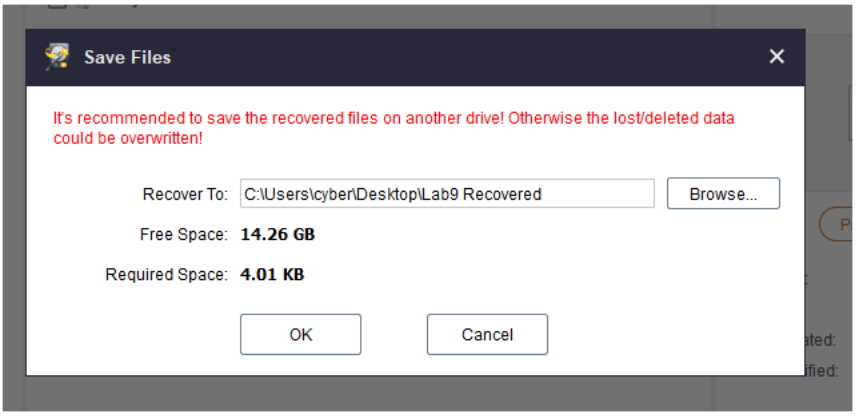
- Open Lab9 Recovered → open PR Test.txt → confirm content reads "Test data added"
- Confirms the file recovered is the exact one stored in the deleted Lab9 Test partition

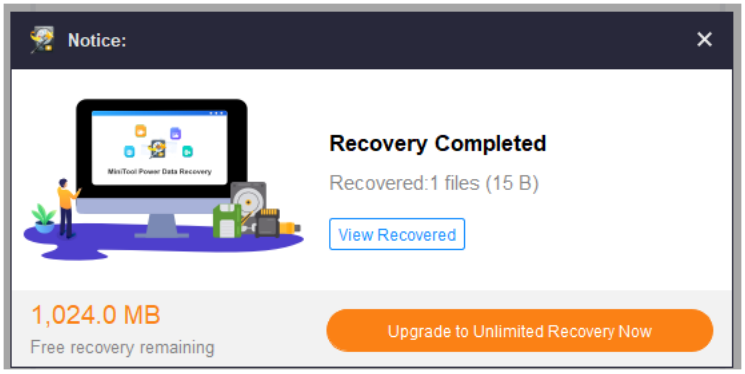
Select Any folder to save the file

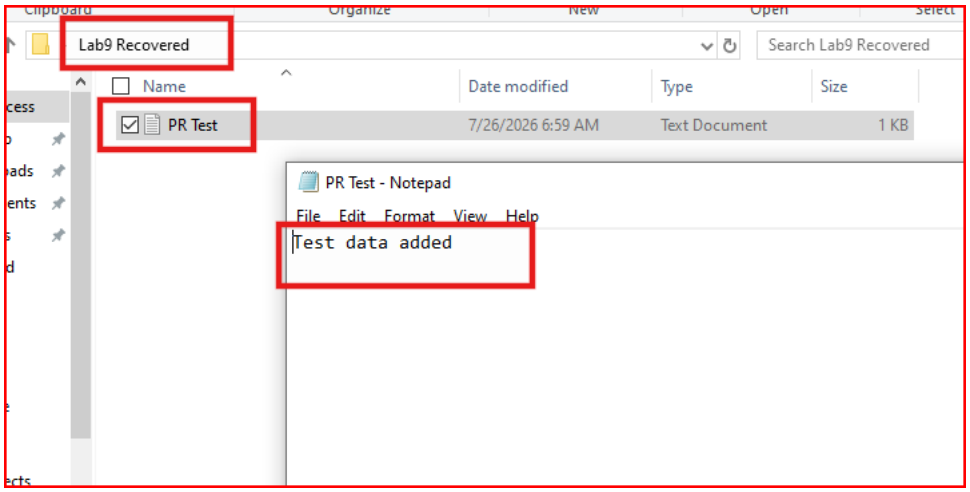
Click on New Folder



Browse the folder where you want to save the file







This file is stored inside the Lab 9 Test partition.

Conclusion

Lab 9 was completed successfully. A file was recovered from a **fully deleted partition** without touching the main Windows system partitions, proving that partition-level deletion leaves recoverable structures until overwritten.

Key Learning Outcome: Create partition → Store test data → Delete partition → Scan lost partition → Locate deleted file → Recover and verify file.

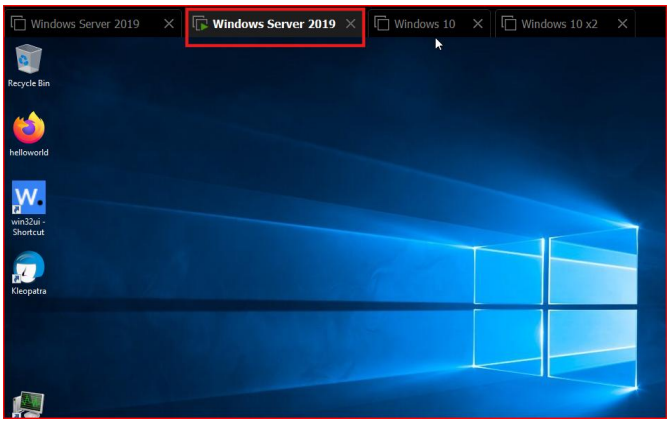
Disk Encryption Using BitLocker

Executive Summary

In this lab you encrypt a separate test drive with Windows BitLocker, protect it with a password, back up a recovery key, verify the drive locks correctly, and confirm it can be unlocked and read again. The Windows **system drive (C:) is never encrypted** — only a dedicated 1 GB test volume.

Item	Details
Virtual machine	Web Server (Server 2, Windows Server 2019)
Tool	BitLocker Drive Encryption
Installation	Not built-in on this server edition — must be added via Server Manager
Encryption target	Safe 1 GB virtual test disk (F:)
Main system drive	Not encrypted
Expected result	Test drive becomes password-protected

Server 2 on

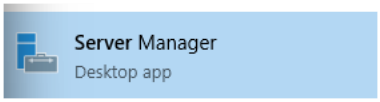


Install the BitLocker Feature

- **Server Manager → Manage → Add Roles and Features → Next to Features**
- **Check BitLocker Drive Encryption → Add Features (pulls in Enhanced Storage, RSAT, BitLocker admin utilities)**
- **Confirm the box remains checked → Next → Install → restart if prompted**

Open Add Roles and Features

1. Open Server Manager if it does not open automatically.

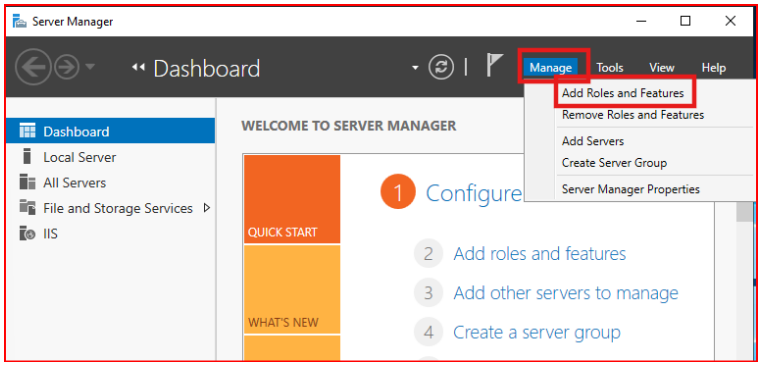


2. In the upper-right corner, click:

Manage

3. **Select:**

Add Roles and Features



Click Next to reach the Features page.

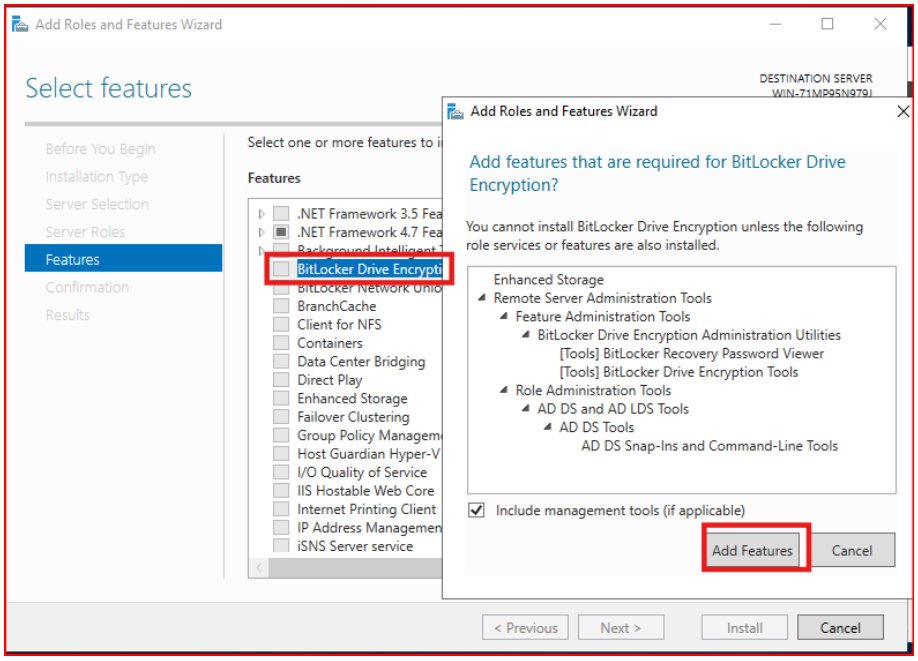
Step 3: Select BitLocker

1. On the Features page, find:

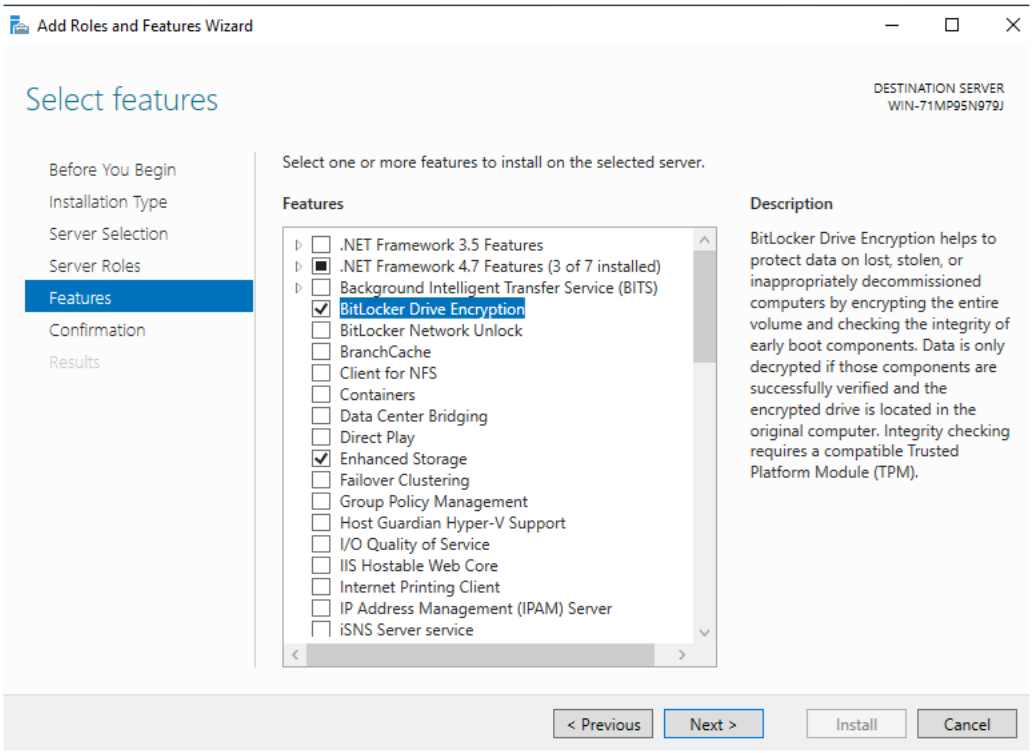
BitLocker Drive Encryption

2. Check its box.
3. When the additional-features window appears, click:

Add Features

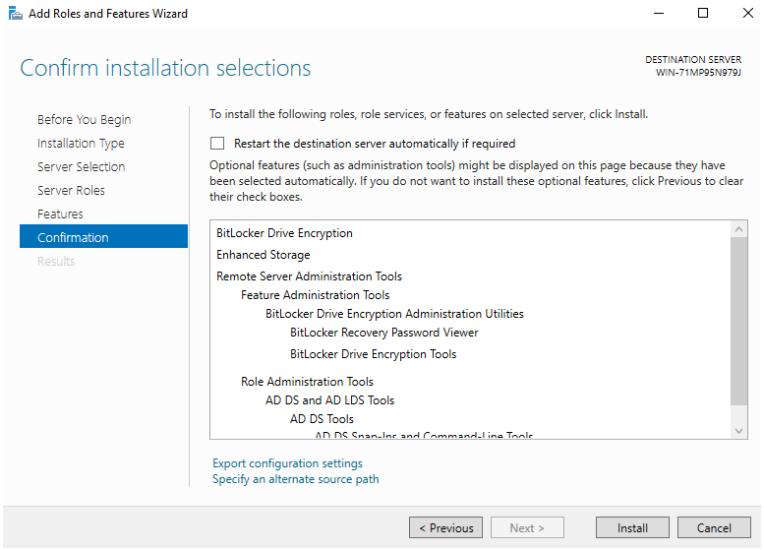


4. Confirm that BitLocker Drive Encryption remains checked.
5. Click Next.

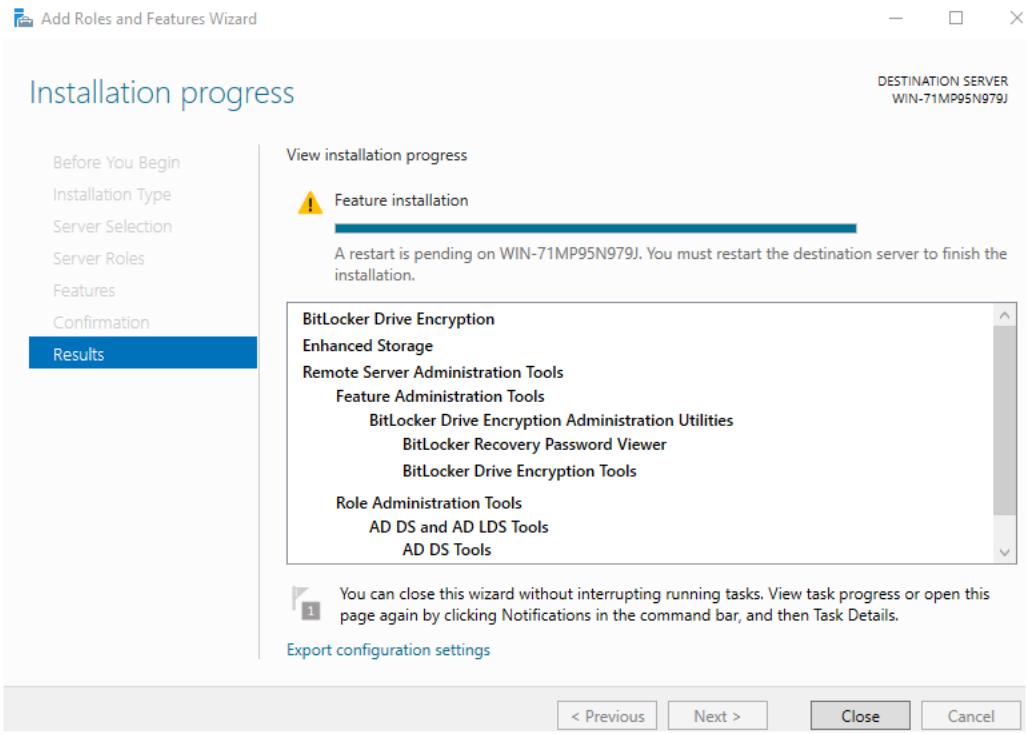


7. **Click:**

Install



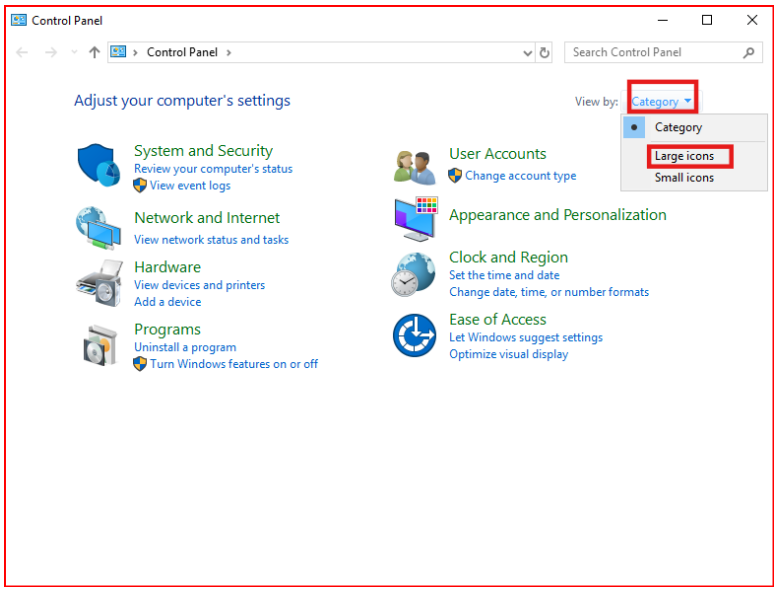
Confirm the restart warning if it appears.



Else restart the machine when installation completes

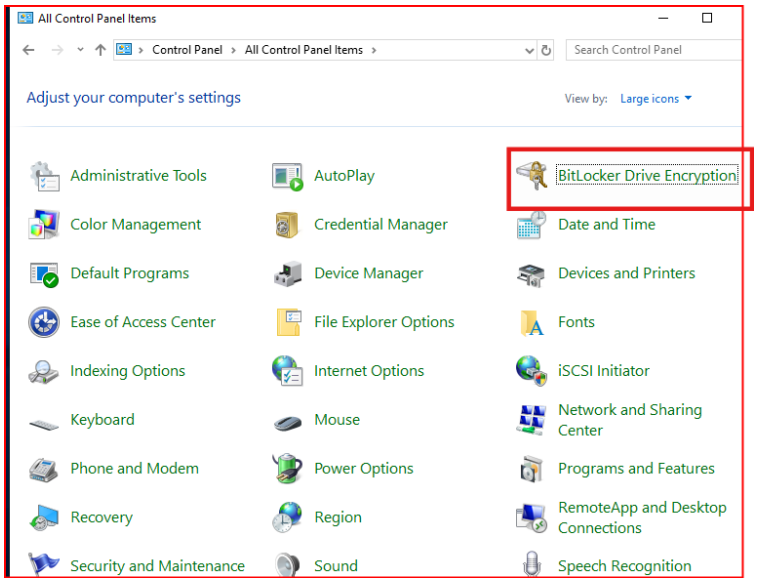
Open Control Pannel

Select Category -> Large Icons



After Restart we have

BitLocker Drive Encryption

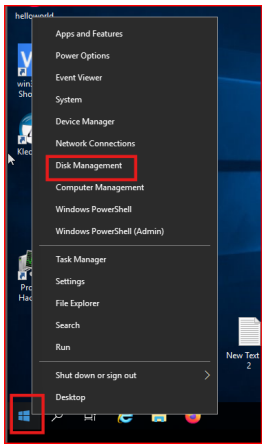


Create a Virtual Test Drive

- Disk Management → Action → Create VHD
- Location: C:\BitLocker-Test.vhdx, Size: 1 GB, Format: VHDX, Type: Dynamically expanding
- New disk shows Unknown / Not Initialized / 1 GB → Initialize Disk → MBR (disk becomes ~1023 MB Unallocated)

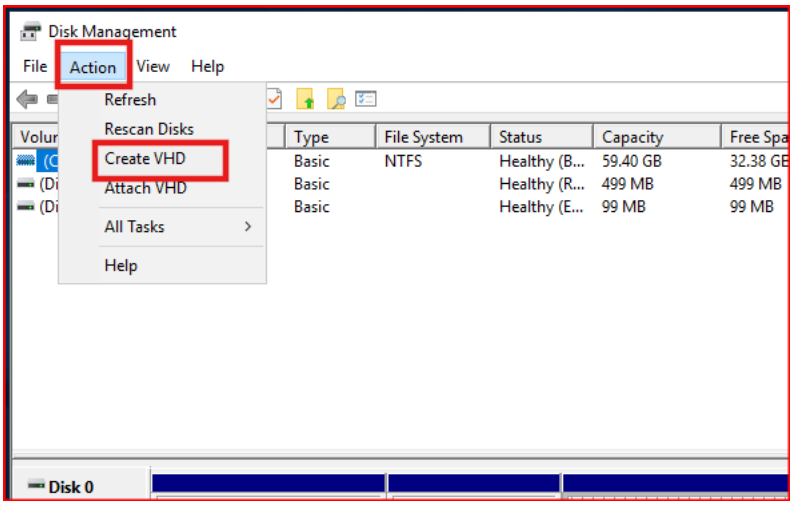
Open Disk Management

1. Right-click the **Start** button.
2. Select: Disk Management



1. In Disk Management, click:

Action → Create VHD



2. Enter these settings:

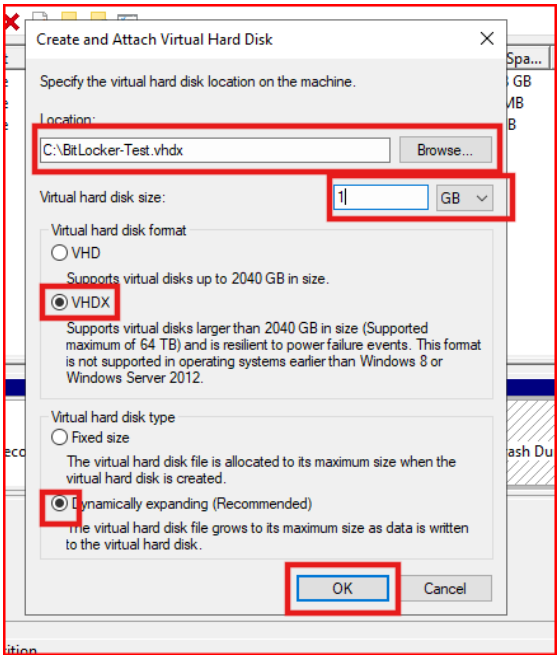
Location: C:\BitLocker-Test.vhdx

Virtual hard disk size: 1 GB

Virtual hard disk format: VHDX

Virtual hard disk type: Dynamically expanding

3. Click **OK**.



4. A new disk should appear as:

Unknown Not Initialized 1 GB



Initialize the New Disk

1. Right-click the left section of the new disk, such as:

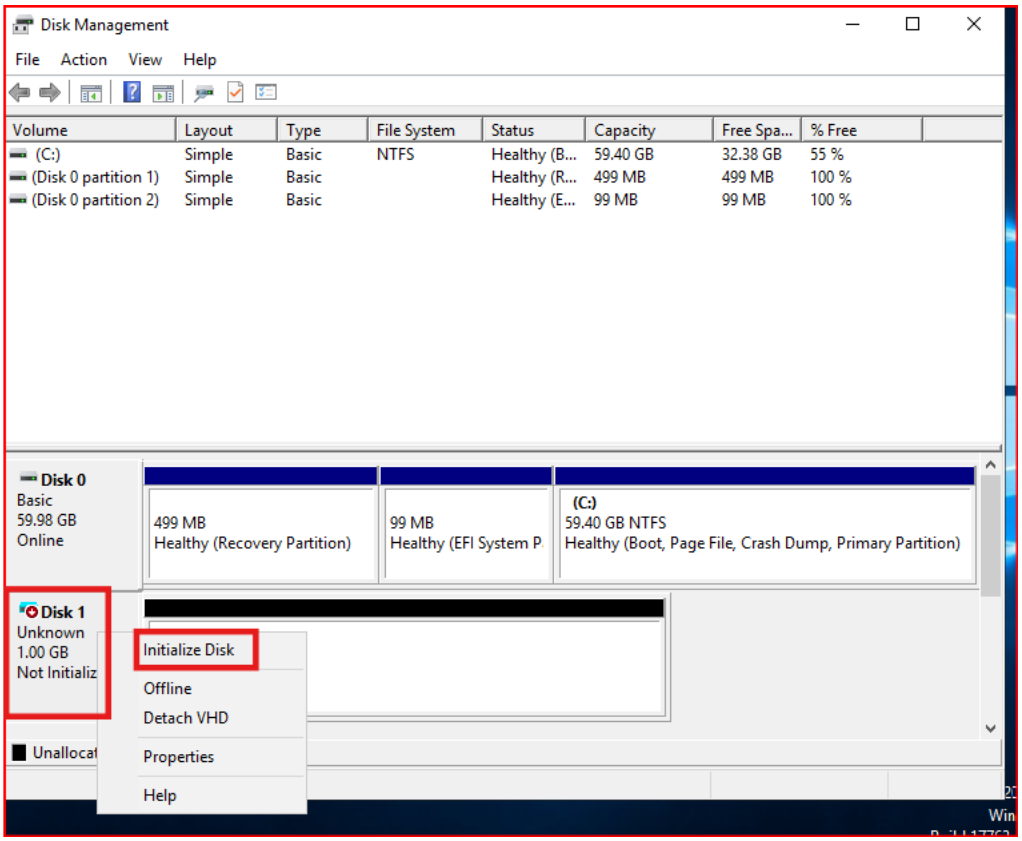
Disk 1

Unknown

Not Initialized

2. Click:

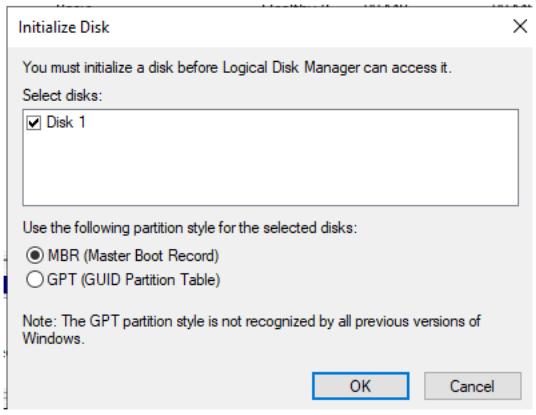
Initialize Disk

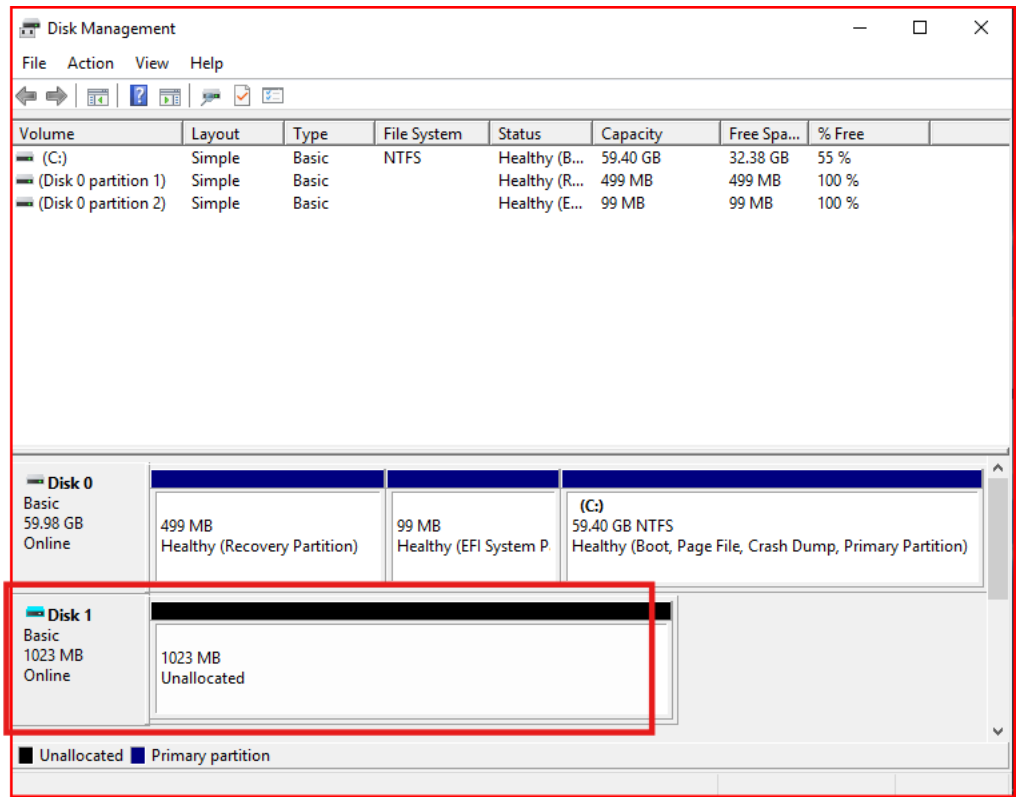


3. Select:

MBR

4. Click **OK**.



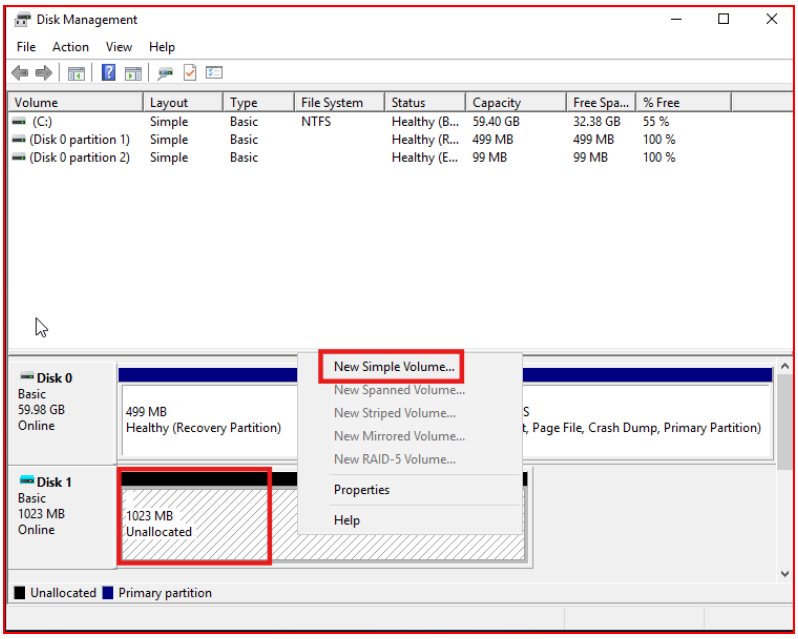


The disk should now display approximately **1 GB Unallocated**.

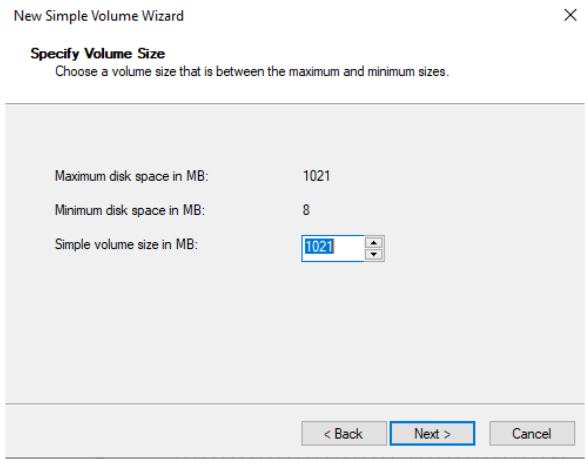
Create and Format the Test Volume

- Right-click the Unallocated space → New Simple Volume → Next → keep max size → Next
- Assign drive letter F: (or next available)
- File system: NTFS, Volume label: BitLocker Test, Quick Format checked → Next → Finish

New Simple Volume



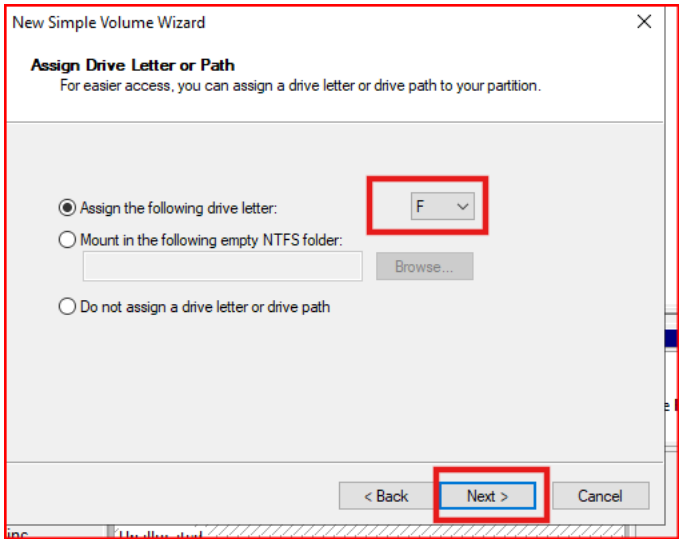
3. Click **Next**.
4. Keep the maximum volume size and click **Next**.



5. Assign:

Drive letter: F:

Use the available letter shown by Windows if F: is already occupied.



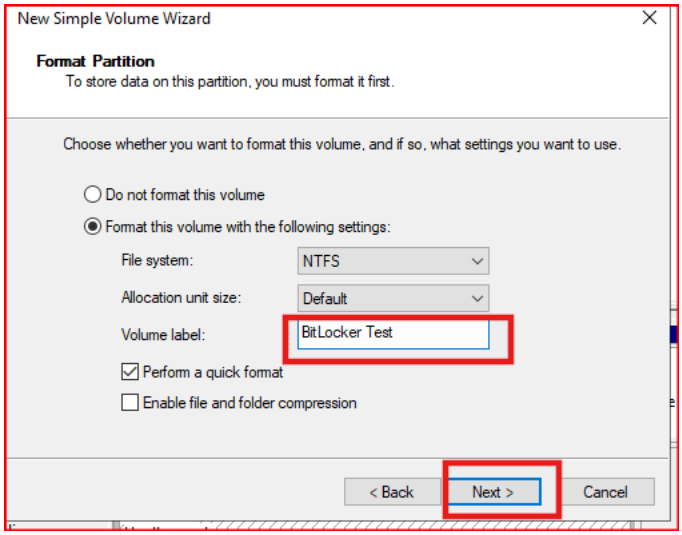
6. Configure:

File system: NTFS

Allocation unit size: Default

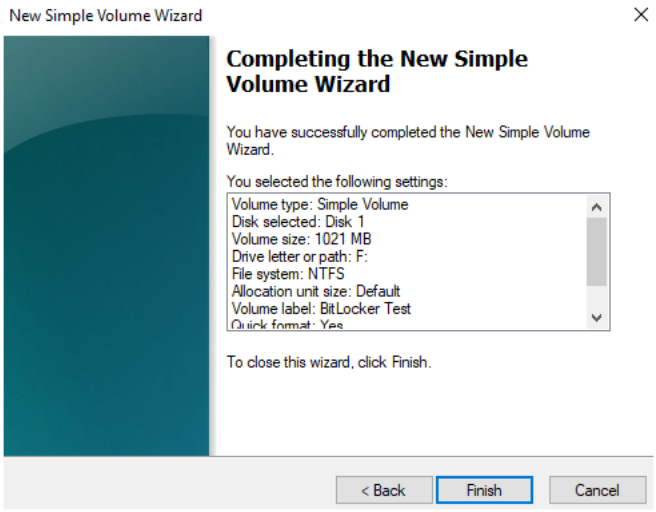
Volume label: BitLocker Test

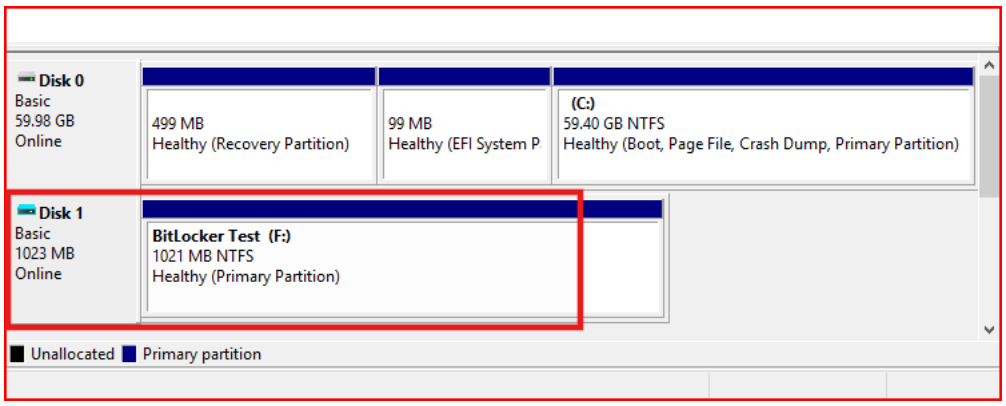
Perform a quick format: Checked



7. Click:

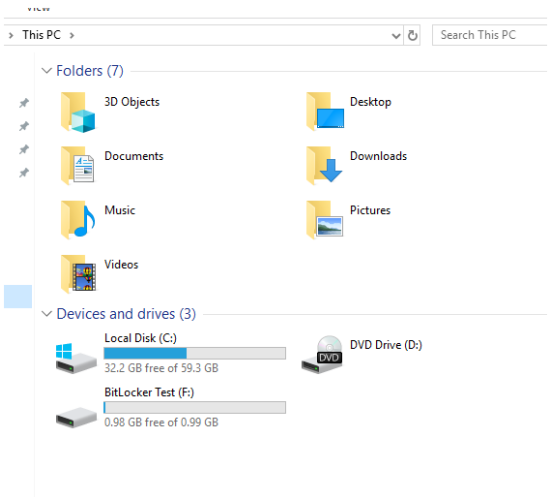
Next → Finish





Create Test Content

- **Open File Explorer → BitLocker Test (F:)**
- **Create Confidential Data.txt, content: "Confidential Data here Be Aware !!!"**
- **Save and close**



BitLocker Test (F:)

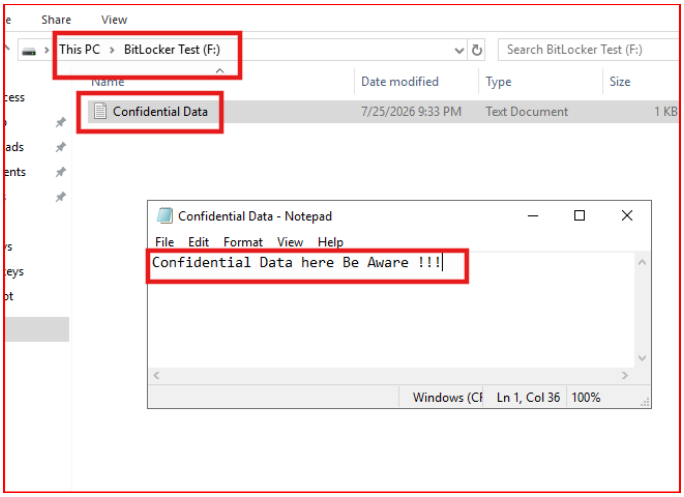
3. Create a text document named:

Confidential Data.txt

4. Open it and type:

This file is protected by BitLocker disk encryption.

5. Save and close the file.



Turn On BitLocker & Set Password

- **Right-click BitLocker Test (F:) → Turn on BitLocker**
- **Select "Use a password to unlock the drive," enter and confirm a strong password (example used: Admin123)**
- **Record the password → Next**

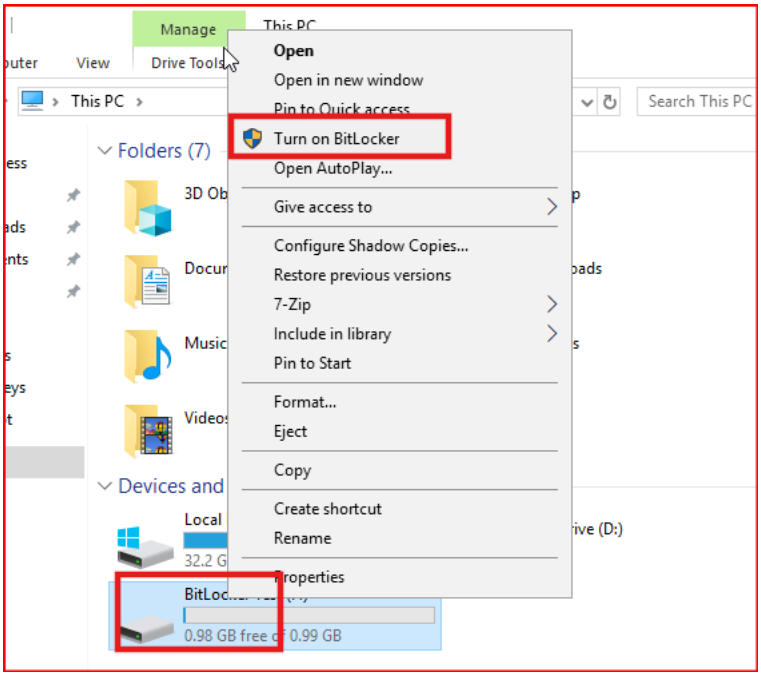
Your **BitLocker Test (F:)** drive and confidential file are ready. ✅

3. Right-click:

BitLocker Test (F:)

4. Select:

Turn on BitLocker



Wait for the BitLocker wizard to open.

1. Set the Unlock Password

1. Select:

Use a password to unlock the drive

2. Enter and confirm a strong password.

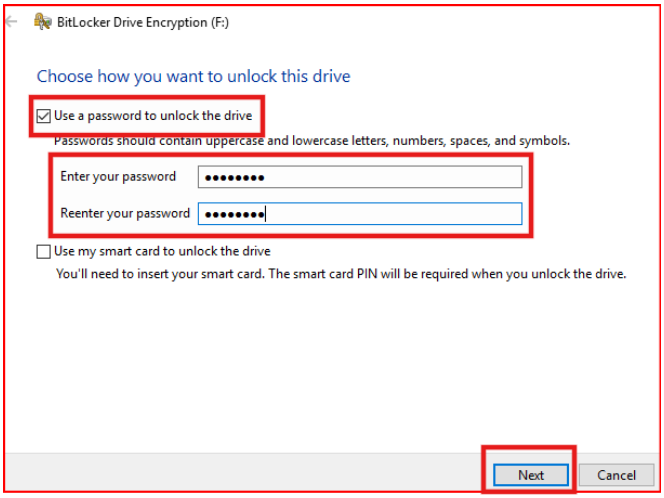
Example for this lab:

Admin123

3. Record the password because you will need it later.

4. Click Next.

The password prevents unauthorized users from opening the encrypted drive.

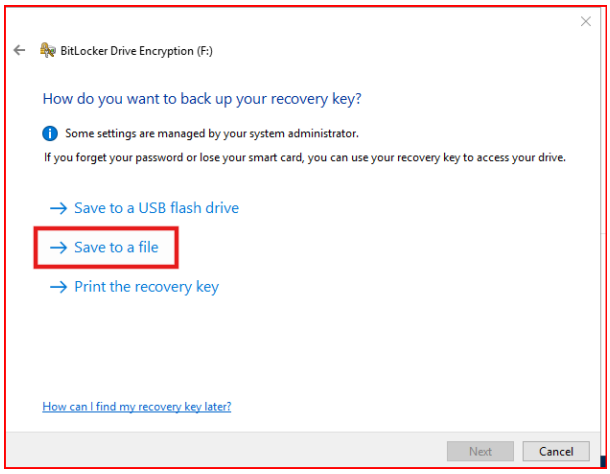


Back Up the Recovery Key

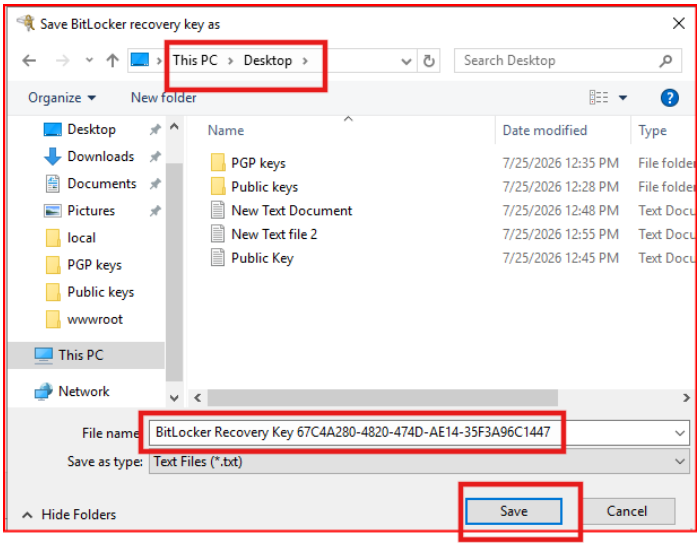
- **Choose "Save to a file," save to the Desktop, keep the default filename**
- **Click Save → Next (recovery key is a secondary unlock method if the password is lost; never store it on the encrypted drive itself)**

Save the Recovery Key

1. **Select:Save to a file**



2. Save the recovery key on the Desktop.
3. Keep the default recovery-key filename.
4. Click Save, then click Next.



The recovery key allows access if the password is forgotten. Do not save it inside the F: drive.

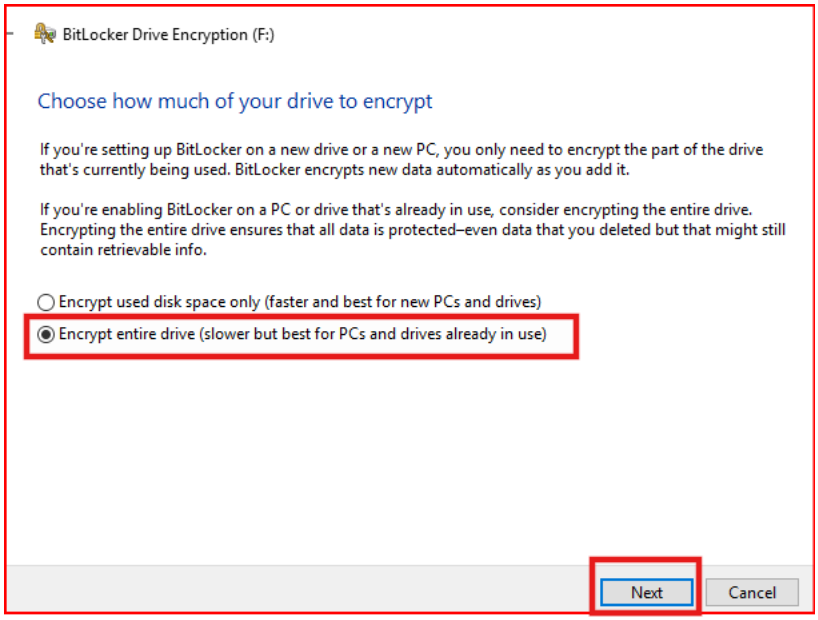
Choose Encryption Scope and Mode

- **Select Encrypt entire drive (covers existing + unused space) → Next**
- **Select New encryption mode (XTS-AES 128) since the drive stays attached to this server → Next**

Encrypt entire drive

Click Next.

This encrypts both the existing data and the unused space on the test drive, matching the lab procedure.



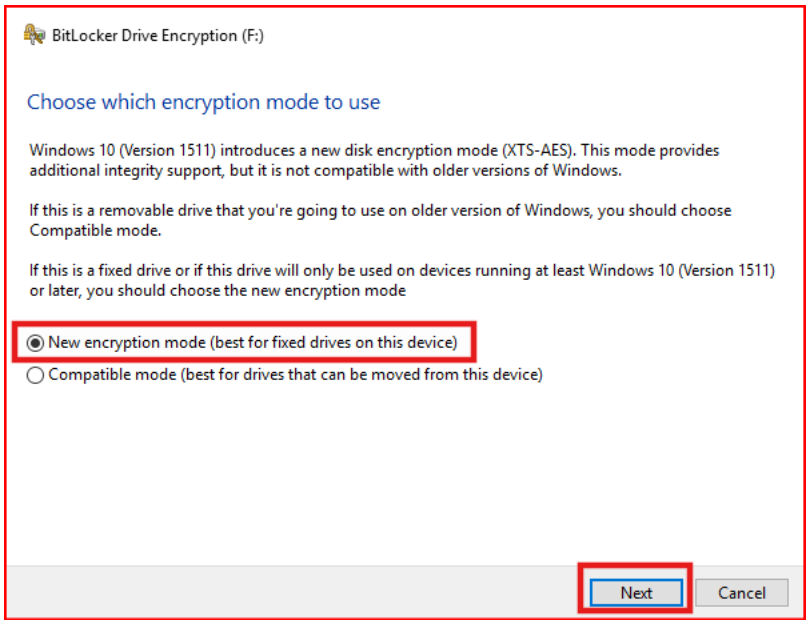
Choose the Encryption Mode

Select:

New Encryption Mode

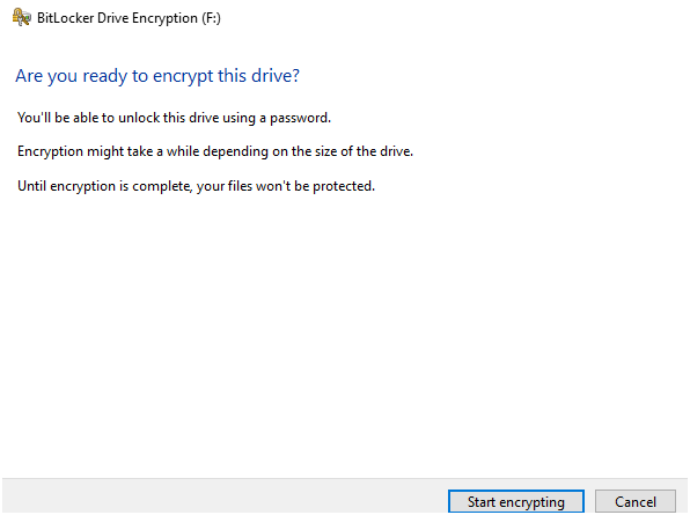
Click Next.

Use this because the virtual test drive will remain attached to this server.



Run Encryption

- **Click Start encrypting, wait until progress reaches 100%, click Close**
- **A padlock icon now appears on the F: drive in File Explorer**



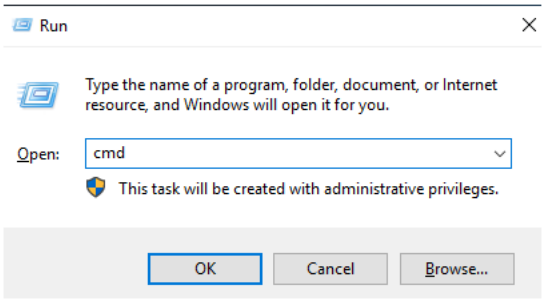
2. Wait until progress reaches 100%.
3. When the message says encryption is complete, click Close.

A padlock icon will appear at the drive we were encrypting



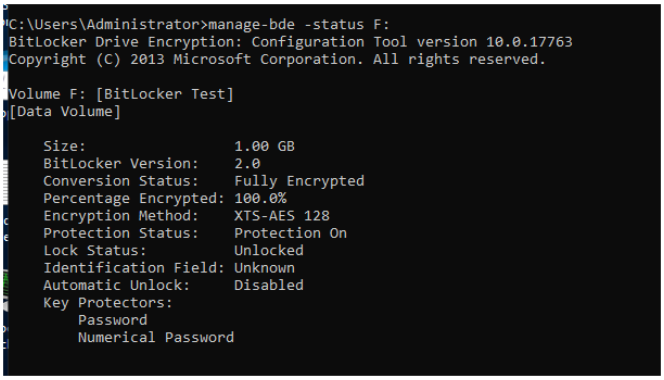
Verify Lock/Unlock via Command Line

- **Close any open files on F:, open Command Prompt as Administrator**
- **manage-bde -status F: → confirms Fully Encrypted, 100.0%, XTS-AES 128, Protection On, Unlocked**
- **manage-bde -lock F: -ForceDismount → confirms "Volume F: is now locked"; padlock icon changes to a locked state**



1. **Check the encryption status:**

manage-bde -status F:

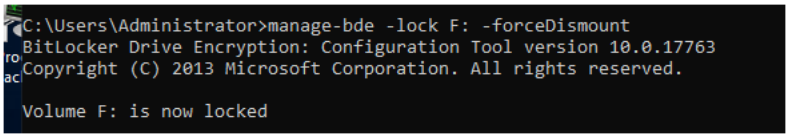


4. **Lock the drive:**

manage-bde -lock F: -ForceDismount

Expected result:

Volume F: is now locked.

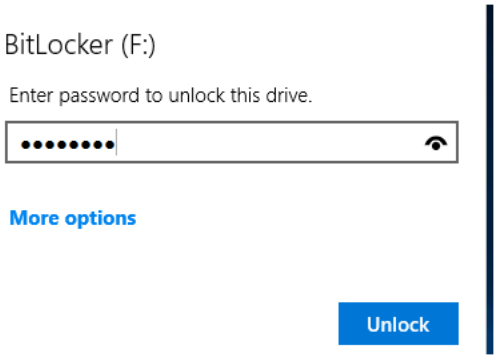


Open This PC. The F: drive should now show a closed padlock.

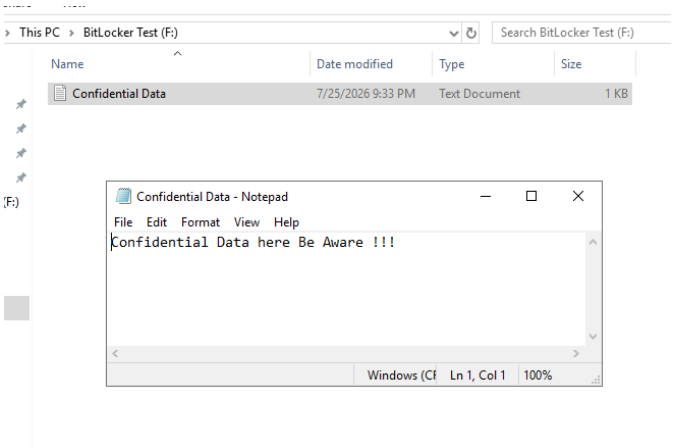


Double-click BitLocker Test (F:).

Enter the BitLocker password and click Unlock.



Open Confidential Data.txt and confirm its content is readable.



Conclusion

Lab 10 successfully demonstrates BitLocker's full lifecycle on a non-system volume: creation, password protection, recovery-key backup, full-drive XTS-AES 128 encryption, forced locking, and password-based unlocking with verified file access. Data on a locked BitLocker volume is completely inaccessible without the password or recovery key.

Key Learning Outcome: Create test drive → Enable BitLocker → Set password → Save recovery key → Encrypt drive → Lock drive → Authenticate → Access protected data.

Data Recovery in Windows Using Disk Drill

Executive Summary

In this final lab you install Disk Drill, permanently delete two of three test items from a separate test volume (reusing the Lab 9 disk), scan the volume, recover the deleted items, and verify their content — demonstrating deep-scan file recovery after Recycle Bin emptying.

Item	Details
Virtual machine	AdminMachine-1
Tool	Disk Drill (v6.3.1398.0, free edition — up to 100 MB recovery)
Test volume	511 MB / ~1023 MB virtual disk reused from Lab 9
Objective	**Recover permanently deleted files**

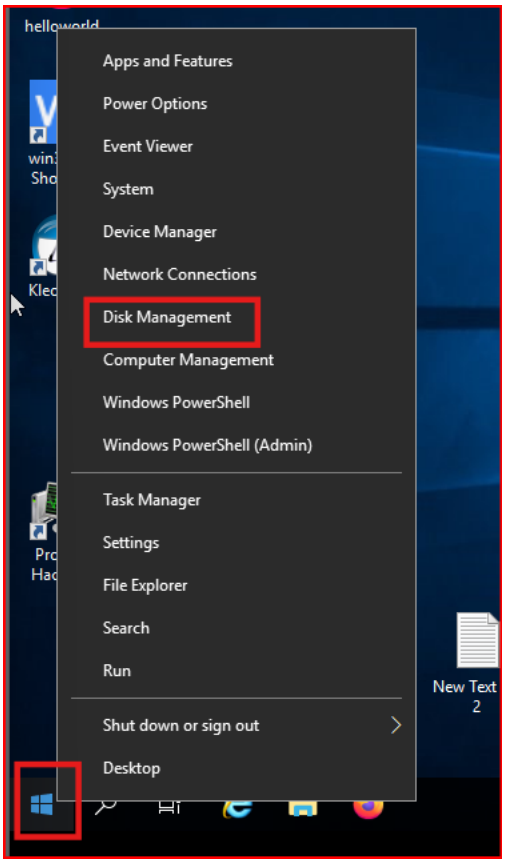
Rebuild the Test Volume

- Disk Management → Action → Create VHD → C:\Lab11-DiskDrill.vhdx, 1 GB, VHDX, dynamically expanding
- Initialize Disk → MBR → disk shows ~1023 MB Unallocated
- Right-click Unallocated → New Simple Volume → assign drive letter (e.g., A:), NTFS, Volume label Lab11 DiskDrill, Quick Format → Finish

Open Disk Management

1. Right-click the **Start** button.
2. Select:

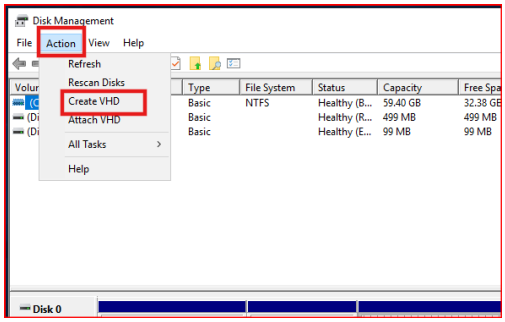
Disk Management



Create a Virtual Hard Disk

2. In Disk Management, click:

Action → Create VHD



3. Enter these settings:

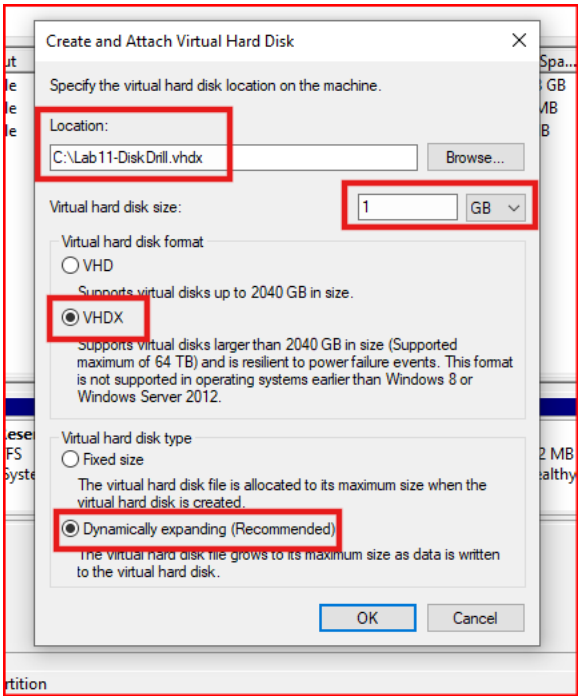
Location: C:\Lab11-DiskDrill.vhdx

Virtual hard disk size: 1 GB

Virtual hard disk format: VHDX

Virtual hard disk type: Dynamically expanding

5. Click **OK**.



6. A new disk should appear as:

Unknown Not Initialized 1 GB



Initialize the New Disk

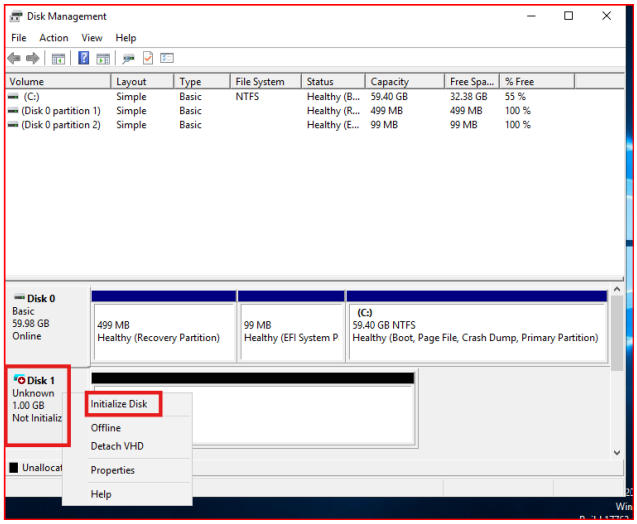
2. Right-click the left section of the new disk, such as:

Disk 1

Unknown

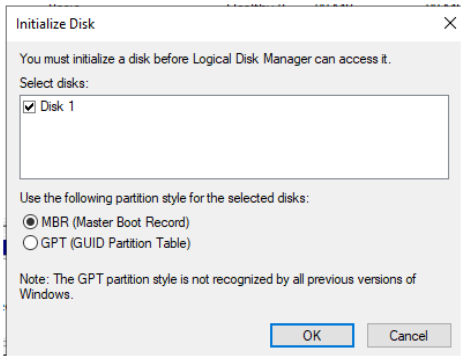
Not Initialized

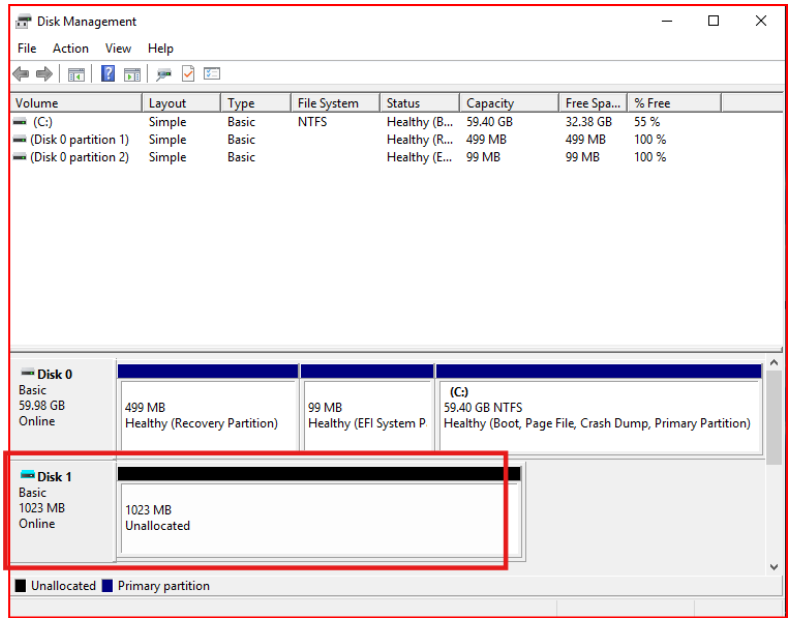
3. Click: Initialize Disk



4. Select: MBR

5. Click **OK**.



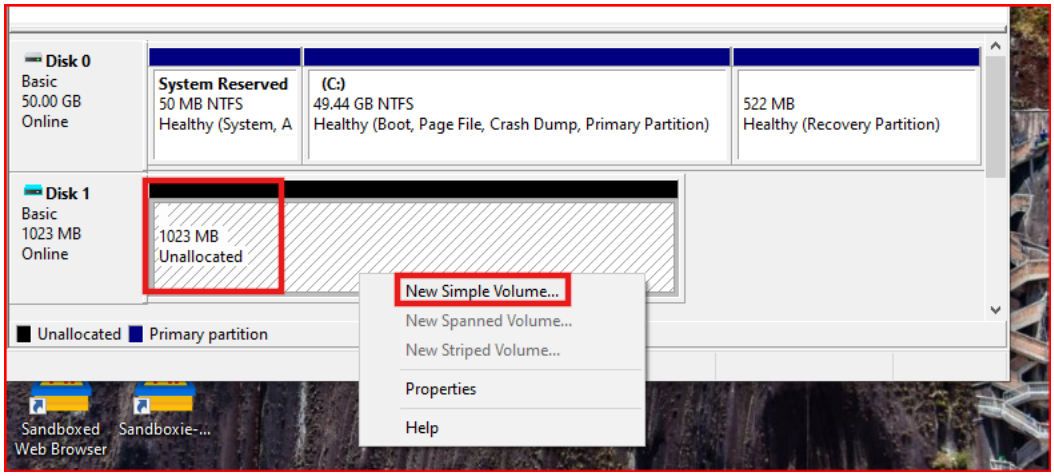


The disk should now display approximately **1 GB Unallocated**.

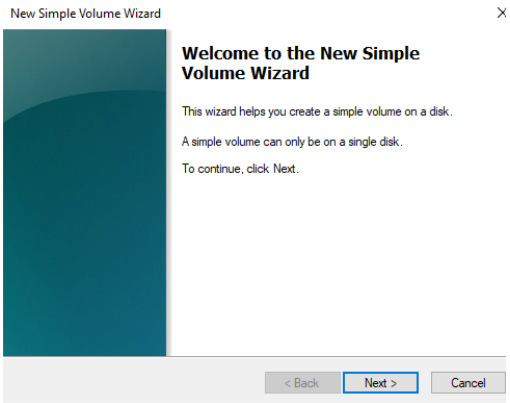
Creating the Test File Structure

- Create folder CND compliance (this one is **not** deleted — stays as a control/baseline)
- Create text file CND file.txt, content: "CND file storage records"
- Create folder CND Ex, containing Example.txt, content: "CND example document"
- Right-click CND Ex → Send to → Compressed (zipped) folder → produces CND Ex.zip
- Delete the original uncompressed CND Ex folder (keep only the zip)
- Drive A: now contains: CND compliance, CND Ex.zip, CND file.txt

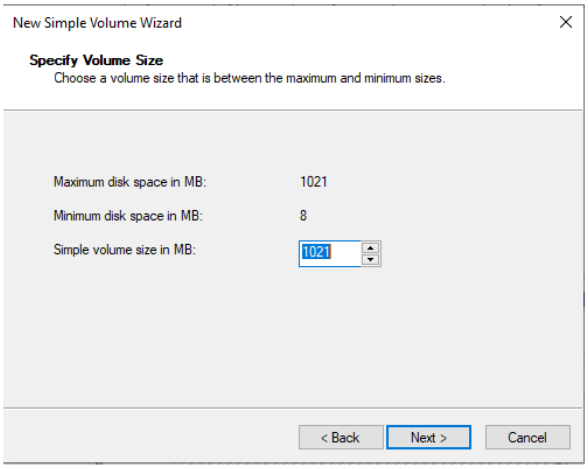
New Simple Volume



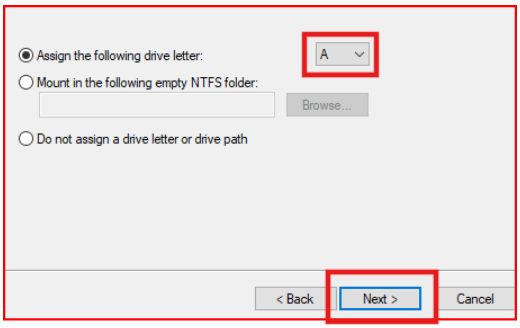
Click next



Click Next (default)



4. Assign an available drive letter, such as: A:



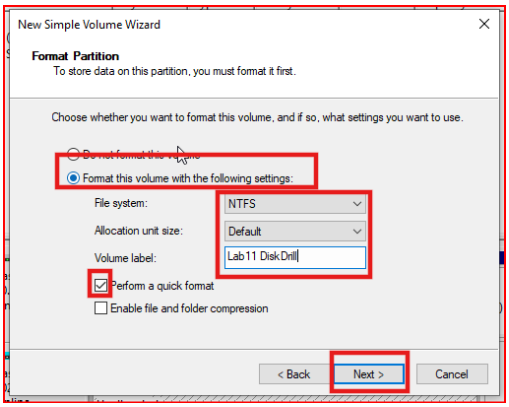
6. Select:

File system: NTFS

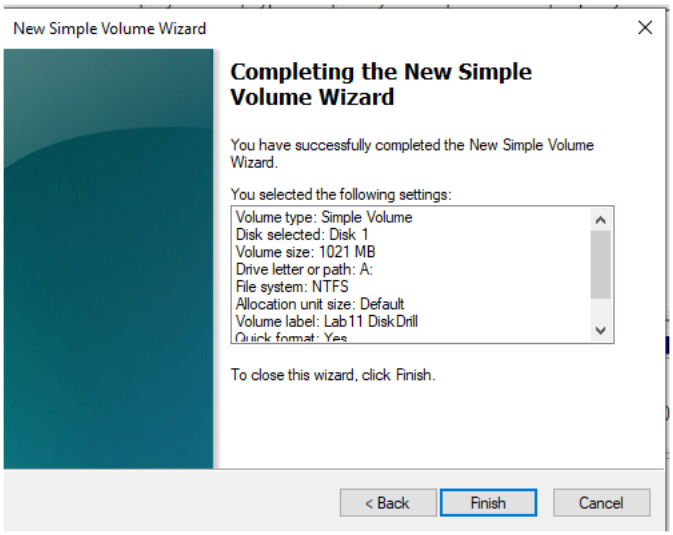
Allocation unit size: Default

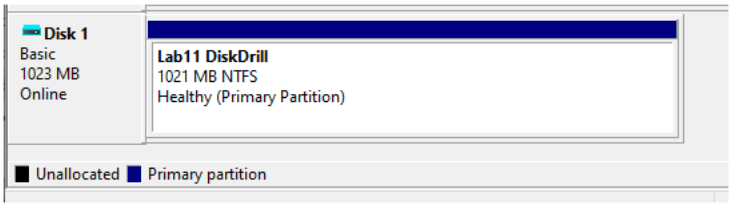
Volume label: Lab11 DiskDrill

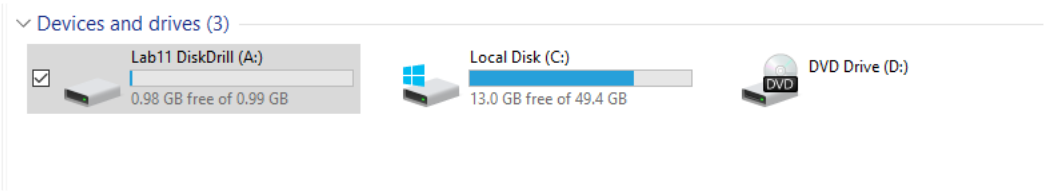
Perform a quick format: Checked



7. Click **Next → Finish**.







Create the Recovery Destination

- On the desktop, create a folder: Lab11 Recovered (recovery target is deliberately off the scanned volume)

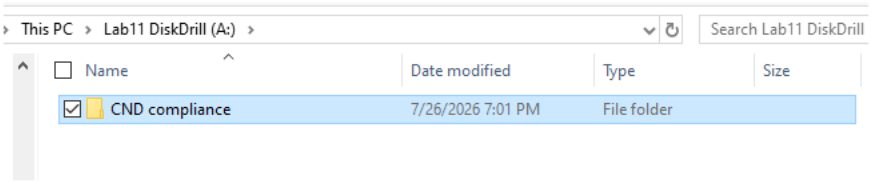


Right-click inside the drive.

Select **New → Folder**.

Name it:

CND compliance



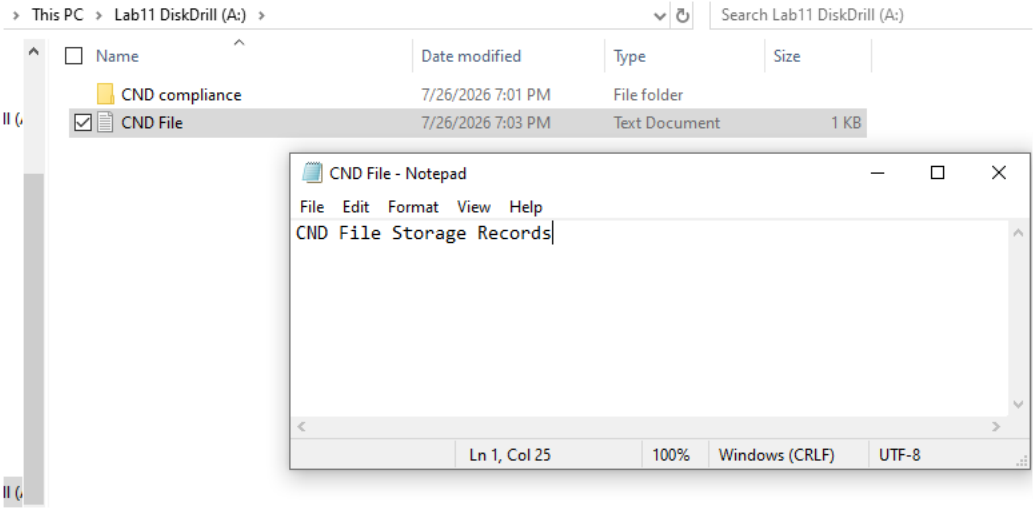
Create the text file that will be deleted

1. Right-click inside the drive.
2. Select **New → Text Document**.
3. Name it:

CND file

4. Open it and type:

CND file storage records

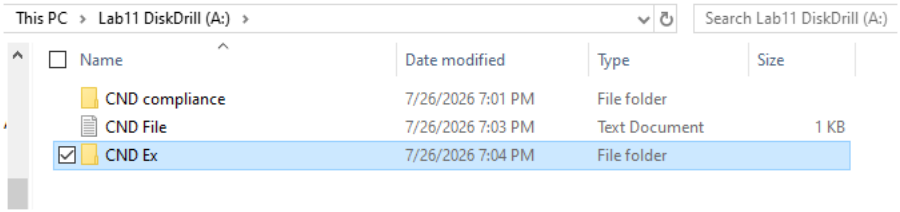


5. Save and close it.

Create the ZIP file that will be deleted

1. Create a folder named:

CND Ex



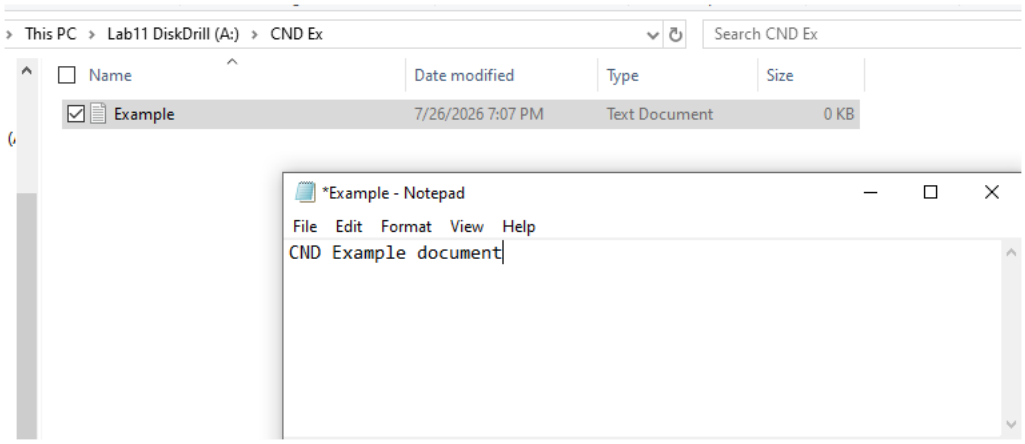
2. Open it and create a text file named:

Example.txt

3. Type:

CND example document

4. Save and close it

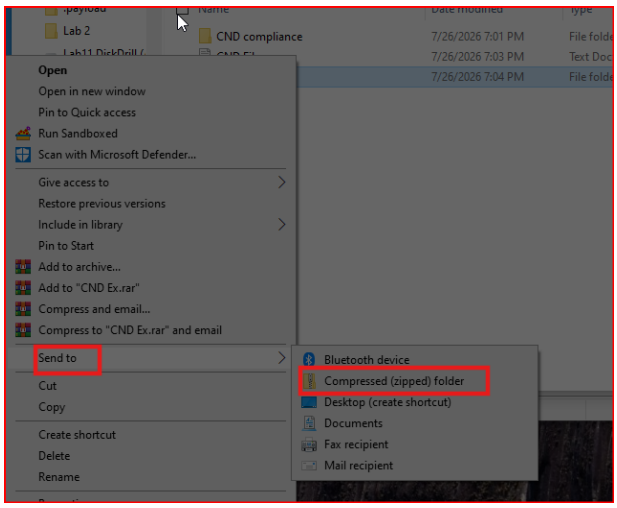


4. Right-click the **CND Ex** folder and select:

Send to → Compressed (zipped) folder

7. Confirm that this file appears:

CND Ex .zip



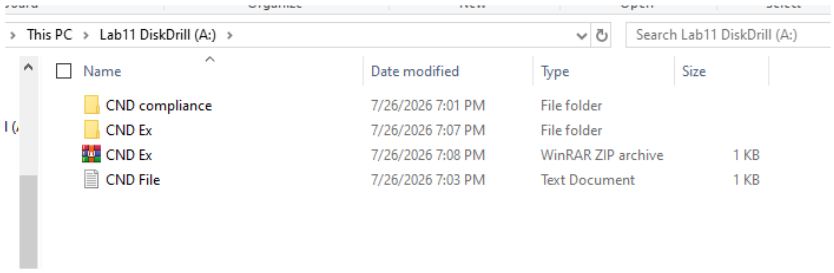
8. Delete only the original uncompressed **CND Ex Document** folder.

The A: drive should now contain:

CND compliance

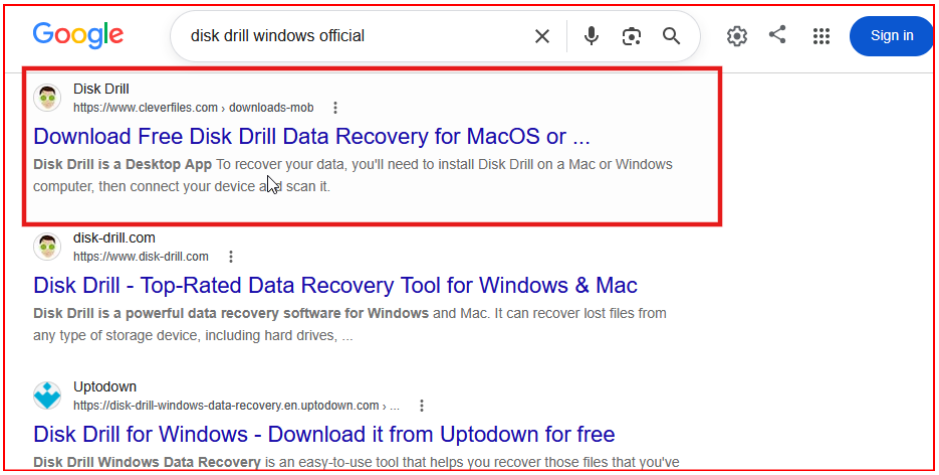
CND file.txt

CND Ex .zip

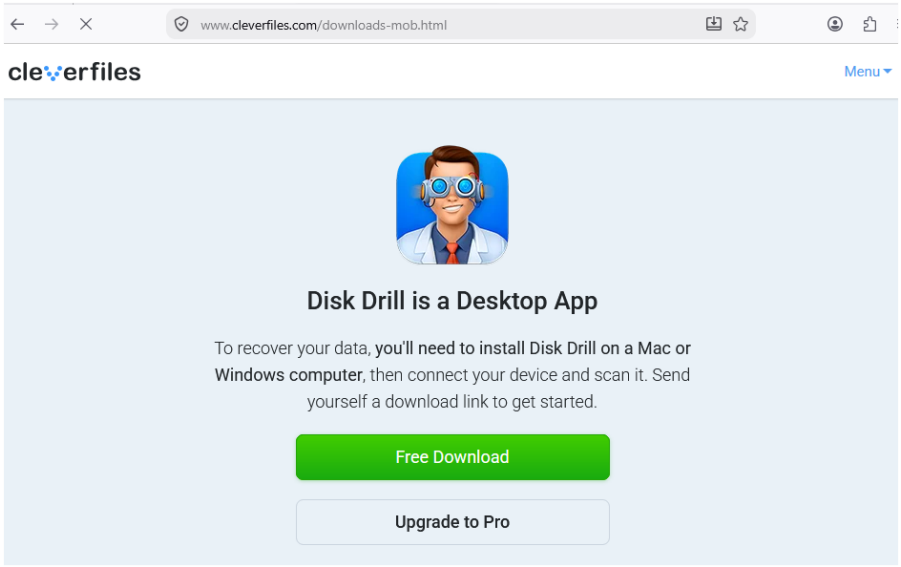


Install Disk Drill

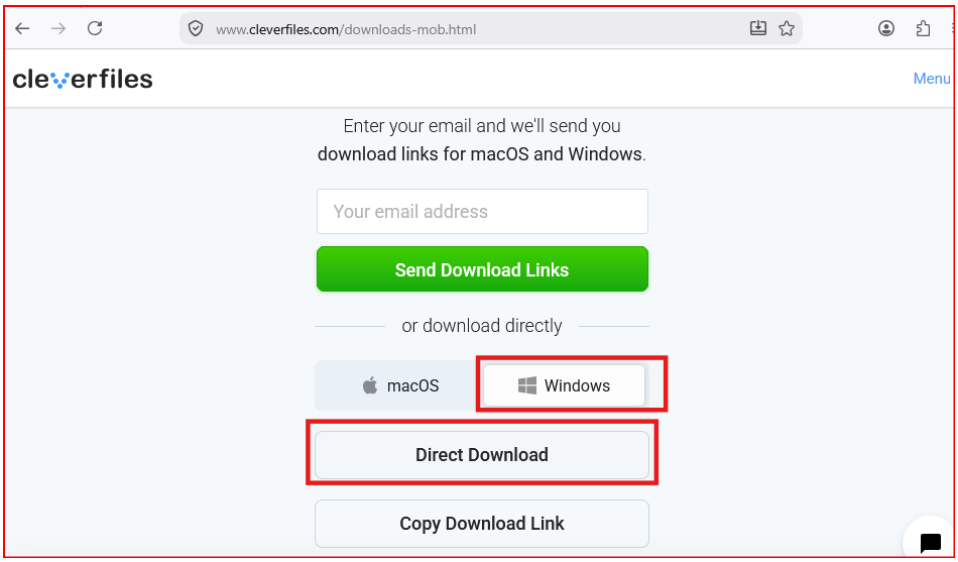
- Search "disk drill windows official," go to the CleverFiles site, click Free Download → Windows → Direct Download
- Run the installer (CleverFiles Disk Drill 6.3.1398.0 Setup) → Install → Launch



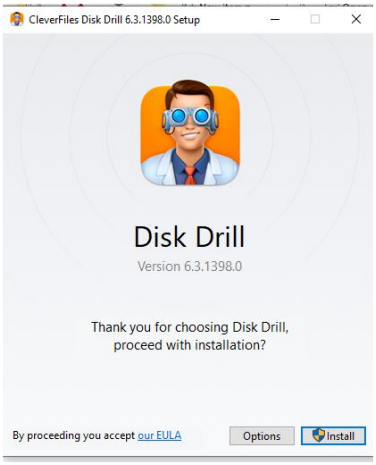
Click on Free Download

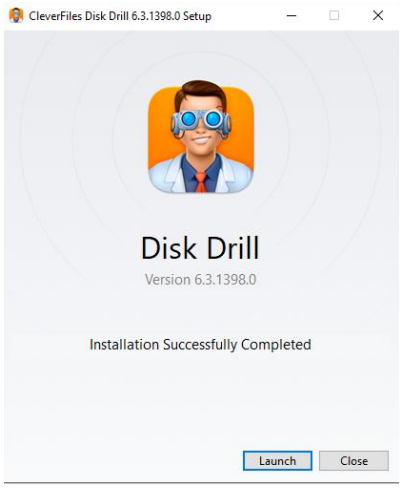


Perform Direct Download

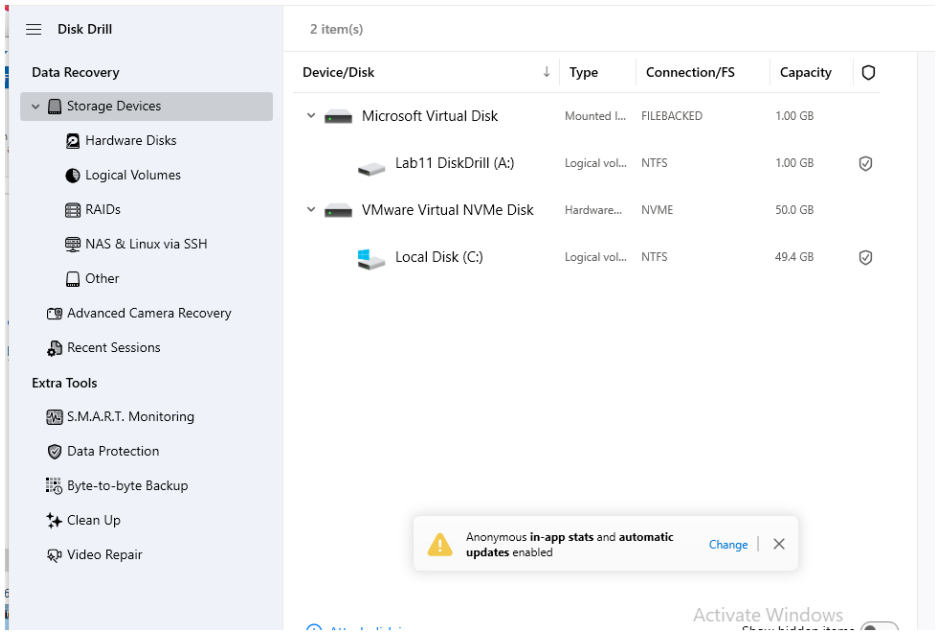


Launch Disk Drill



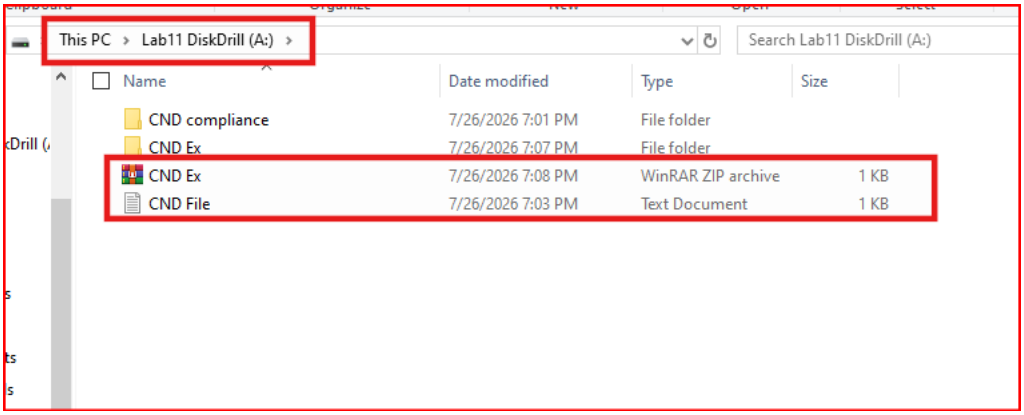


Disk Drill Ready

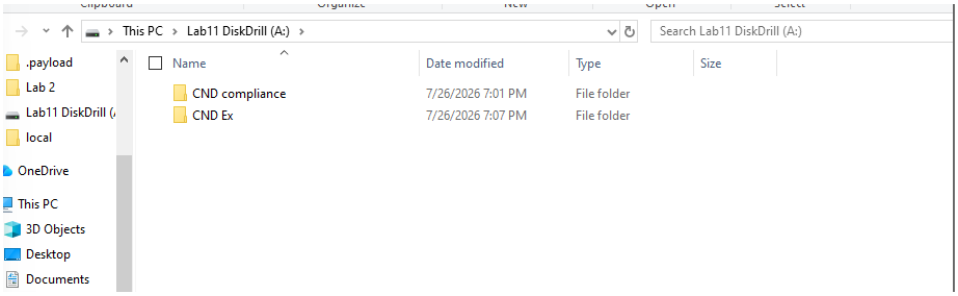


Permanently Delete the Test Files

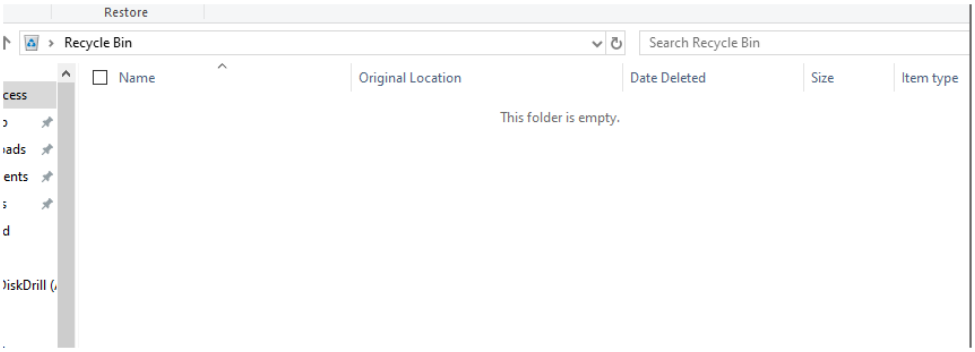
- Open Lab11 DiskDrill (A:), select CND file.txt and CND Ex.zip, delete both
- Empty the Recycle Bin and confirm it's empty (only CND compliance remains on A:)



Delete both of them

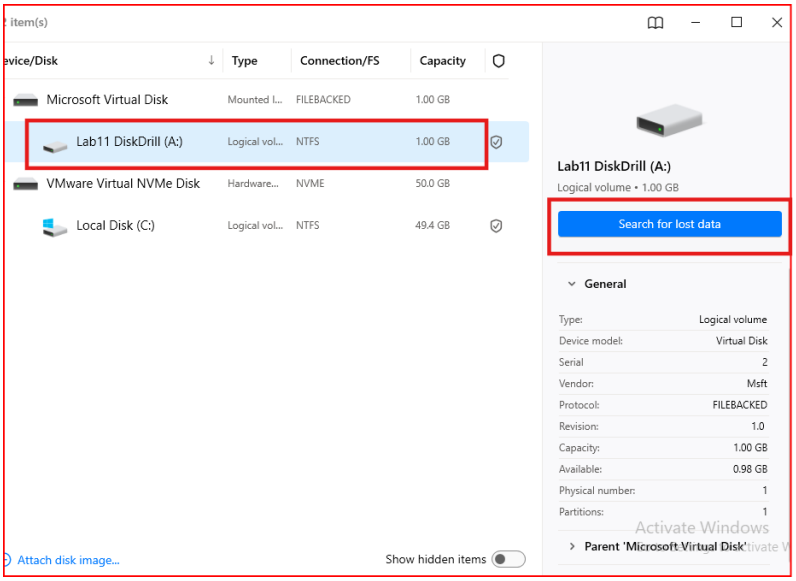


Clear Recycle bin it should be empty too

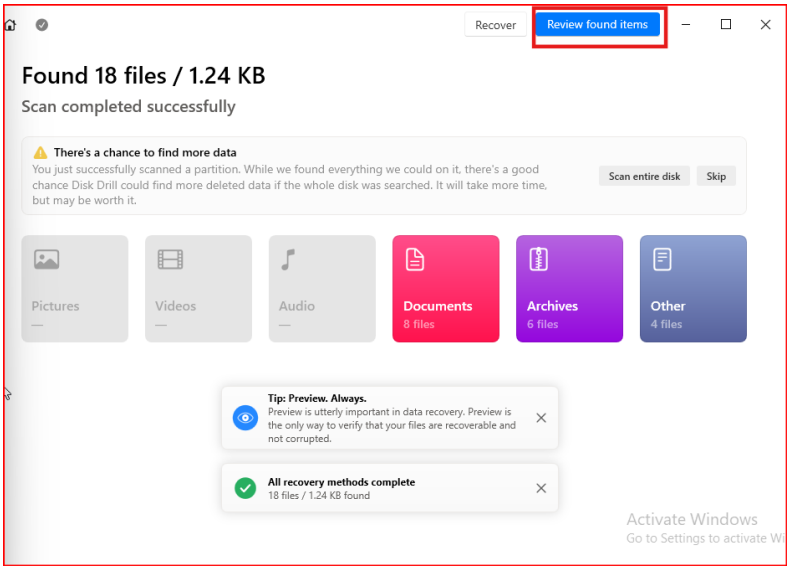


Scan the Test Volume

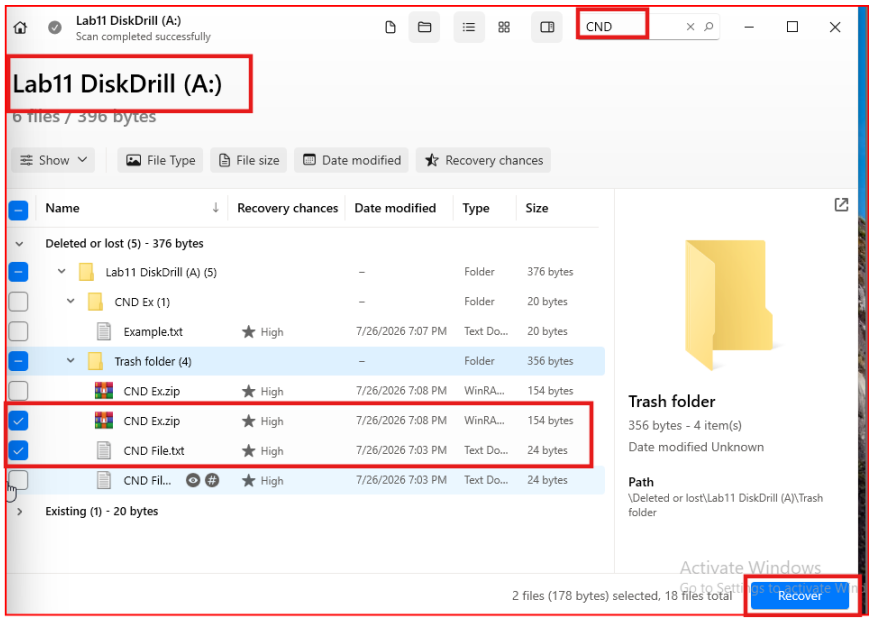
- Open Disk Drill → locate Lab11 DiskDrill (A:) under Storage Devices → select it → click **Search for lost data**
- Scan completes: **18 files / 1.24 KB found** — Documents: 8, Archives: 6, Other: 4 ("All recovery methods complete")



5. Allow the scan to complete.

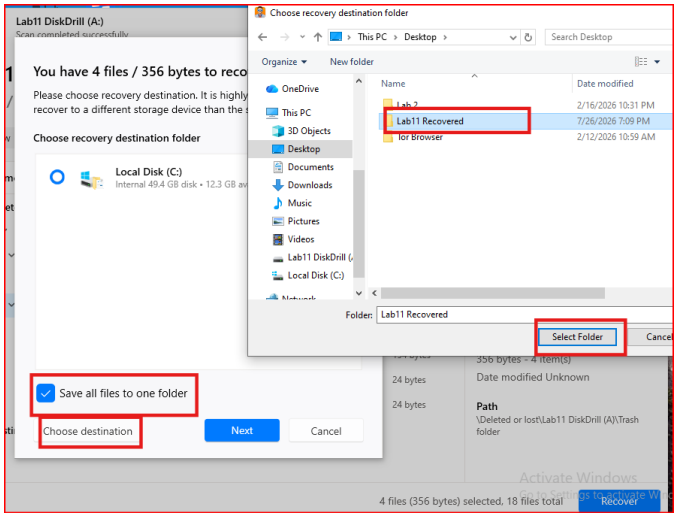


6. Search for: CND
7. Locate:
    - CND file .txt
    - CND Ex.zip
8. Select both files.
9. Preview the text file when the preview option is available.



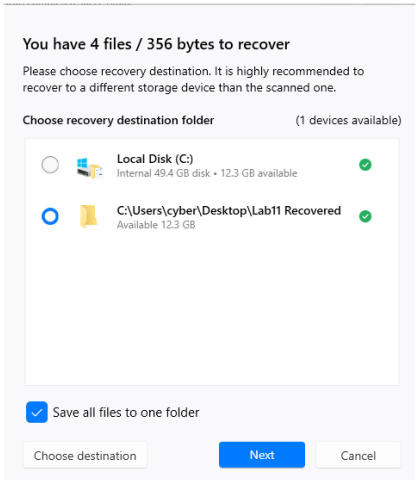
Recover the Files

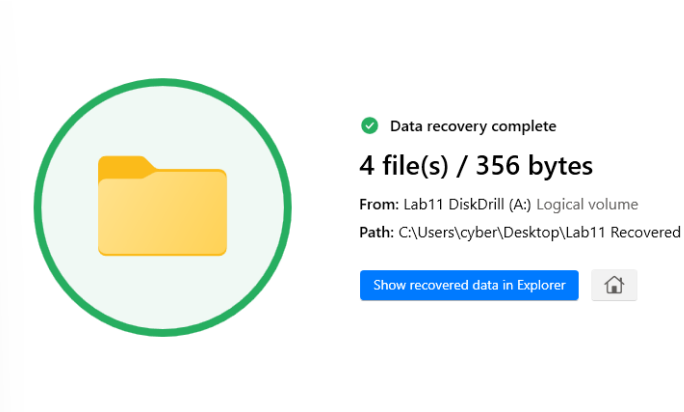
- Click **Recover**
- Choose destination: Desktop → Lab11 Recovered (a second storage location, per Disk Drill's recommendation)
- Confirm "You have 4 files / 356 bytes to recover" (includes duplicate entries) → Next → recovery runs
- Result: "Data recovery complete — 4 file(s) / 356 bytes recovered from Lab11 DiskDrill (A:) → C:\Users\cyber\Desktop\Lab11 Recovered"



3. Start the recovery.

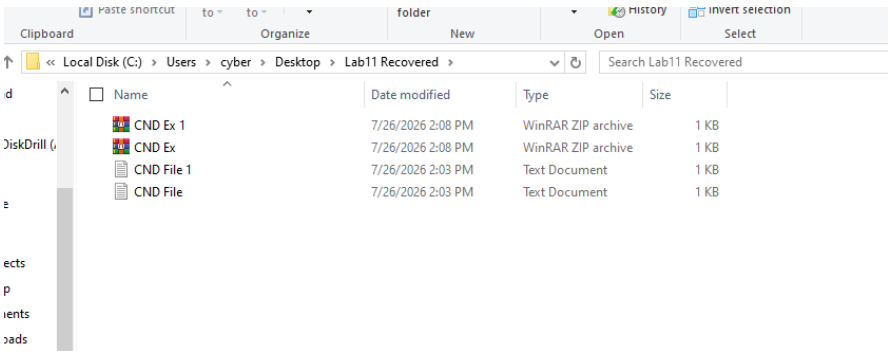
   Click Next



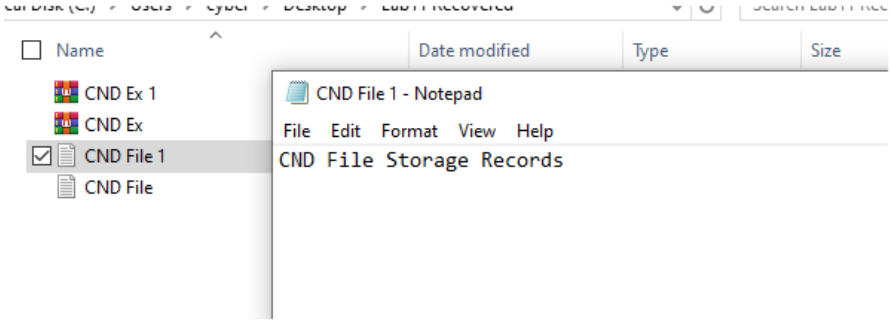


4. Open the **Lab11 Recovered** folder.
5. Confirm:

Data Recovered



Data Check



CONCLUSION

Reactive capability (Labs 8, 9, 11): All three recovery tools successfully located deleted data down to matching filenames, timestamps, and content — proving that "deleted" is not "destroyed." Only Lab 8 stopped short of a full save, due to a licensing restriction rather than a technical failure of detection.

Preventive capability (Lab 10): BitLocker showed that encryption is what actually blocks unauthorized access a locked, encrypted volume is unreadable without the password or recovery key, regardless of physical access to the disk.

Combined takeaway: Recoverability and encryption are two sides of the same data-governance coin. Recovery tools protect against accidental loss; encryption protects against unauthorized access. A resilient environment needs both — encrypting sensitive volumes while maintaining tested recovery procedures and secure recovery-key storage.